

Canonical memory, colliding polynomial filters and original observed energy

Split-Zero research programme

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This continuation evaluates the complete memory of the original canonical arithmetic action below the kq scale, including zero regularizer. The proof retains the original kernel, every invariant column, the full source minimum and the physical coordinate $S = k/2 + iy$:

$$0 \leq g_N(z) \leq g_N(0) = O_h(q \log(q/k)) = o_h(kq), \quad z \geq 0.$$

The original energy quotient and the entire energy graph also have zero four-cutoff coefficient at scale kq . Exact determinant identities transfer these values into the earlier kernel and observed calculations. For positive regularizers with $|\log z_N| = o(q)$, the original observed compression and full action have the same return through $o(kq)$.

A second calculation covers monic filters p_k of degree d_k whose roots lie in a fixed compact set disjoint from the original odd lattice. Every collision and multiplicity is retained. For $d_k \log(q+2) = o(q)$, the complete original observed rational-source determinant has return $C_{\partial d_k q} + o(d_k q)$, and exactly d_k slow singular directions are determined. The explicit supplied filter

$$p_k(y) = [(y-13)(y-4)(y-18)(y-468)]^k$$

therefore gives $4C_{\partial kq} + o(kq)$, with $5.415572 < 4C_{\partial} < 5.419004$.

The finite reduced resolvent and heat estimates include all remaining modes. Repeated slow poles retain their full observed mass matrix. Every slow eigenvalue, including multiplicities, transfers to the actual energy graph with an explicit relative error; the whole slow cluster receives the evaluated determinant coefficient.

The proofs below state all finite guards. Endpoint coefficients retain the scope and proof status of the earlier monic-profile and zero-law inputs. The initial canonical kernel coefficient and original complex current phases still require their own calculation. Human sources are cited where used; complete prior programme providers accompany this edition, with their original attribution.

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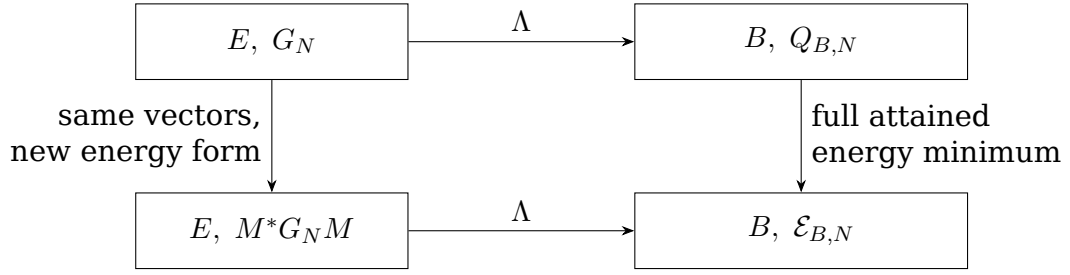
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The exact maps and determinant changes

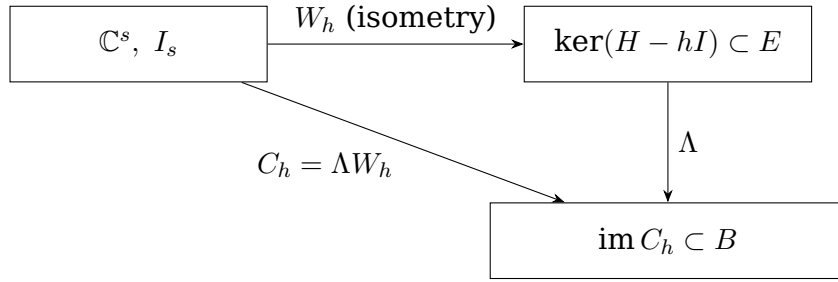


Kernel factor: $\det A_N$
 Observed factor: $|\det M|^2 / \det A_N$

$$\mathcal{R} \log \det A_N = o(kq)$$

$$\mathcal{R} \log [\det \mathcal{E}_{B,N} / \det Q_{B,N}] = o(kq)$$

Figure 1: The same original observation with its two fully attained metrics. The vertical arrows state a change of positive form on the same vector spaces; they are not asserted to be isometries. BR1–5 proves both determinant factors and their cancellation. ME29–34 evaluates the finite gap and the resulting $o(kq)$ error.



$$\Theta_h = C_h^\dagger C_h > 0$$

$$Y_h = C_h \Theta_h^{-1/2} \text{ is an isometry}$$

$$Y_h^\dagger \mathcal{F}'(-h) Y_h = \Theta_h^{-1}$$

$$\text{Res}_{z=-h} \mathcal{F}(z)^{-1} = Y_h \Theta_h Y_h^\dagger$$

Figure 2: A repeated eigenvalue of multiplicity s retains an entire positive matrix of observed masses. The domain and target metrics are stated. RM24–26 proves reconstruction, the derivative and the residue, without diagonalizing the observed mass by assumption. The Feshbach–Schur source is Dusson, Sigal and Stamm (2021), original-author theorem `thm:isospF` and equation `QP`; the full finite argument appears in this reader.

Part I

Original programme receivers

Receiving the memory and filter calculations in the original programme

These are complete receiving proofs for SZ-20260921-016. Keep the original quartet, divisor, source mass, physical coordinate $S = k/2 + iy$, actual period and full kernel frame of ME1-6. Every return below uses the four original cutoffs $q-1, q, 2q-1, 2q$ with signs $+, +, -, -$. Write this linear operation as $\mathcal{R}$. Constants implicit in the stated asymptotics have the fixed original data specified in ME and RF. All finite domains in those proofs remain in force.

1. The original kernel receives an evaluated energy-action correction

Let $I_K : \mathbb{C}^m \rightarrow E$ be the original full coefficient frame, $H_{K,N} = I_K^* G_N I_K$, and let M be the original multiplication by y . It is invertible because all original roots are nonzero. The image MI_K is therefore a full frame of MK . Put

$$H_{K,N}^{\text{en}} = I_K^* M^* G_N M I_K. \quad (\text{BR1})$$

The map $I_K H_{K,N}^{-1/2}$ is an isometry onto the original kernel. Thus the block A_N in ME7 is exactly

$$A_N = H_{K,N}^{-1/2} H_{K,N}^{\text{en}} H_{K,N}^{-1/2}, \quad \log \det H_{K,N}^{\text{en}} - \log \det H_{K,N} = \log \det A_N. \quad (\text{BR2})$$

This proves the full change of metric, including its determinant factor. ME32-34 now gives

$$|\mathcal{R} \log \det H_{K,N}^{\text{en}} - \mathcal{R} \log \det H_{K,N}| \leq \sum_N |\log \det A_N| = O_{h,A}(q \log(q+2)) = o(kq). \quad (\text{BR3})$$

The same formula describes the original source norm on the actual image MK , through the map $M|_K$. It evaluates the action-induced change of the initial kernel allocation. The initial allocation itself remains the full original determinant, with all invariant columns retained.

For the observed metric $Q_{B,N}$, the actual energy quotient is

$$\mathcal{E}_{B,N} = [\Lambda(M^* G_N M)^{-1} \Lambda^*]^{-1}. \quad (\text{BR4})$$

ME33 proves

$$\log \det \mathcal{E}_{B,N} - \log \det Q_{B,N} = 2 \log |\det M| - \log \det A_N. \quad (\text{BR5})$$

The first term is independent of N , so it cancels under $\mathcal{R}$. Therefore the observed change is exactly the negative of the kernel change, and both are $o(kq)$. These are complementary restrictions and attained quotients of the same original energy form. They have been connected by the complete determinant factorization, rather than by a rank allocation.

2. The graph correction has its own evaluated determinant

In original isometric coordinates, retain $L_{\text{en},N} = (-A_N^{-1}B_{H,N}, I)^T$, $Z_N = L_{\text{en},N}^\dagger L_{\text{en},N}$, and $J_{\text{en},N} = L_{\text{en},N} Z_N^{-1/2}$. ME31–33 proves the exact relation

$$\log \det(J_{\text{en},N}^\dagger H_N J_{\text{en},N}) = 2 \log |\det M| - \log \det A_N - \log \det Z_N. \quad (\text{BR6})$$

Consequently ME34 gives a zero kq coefficient for the four-cutoff return of the whole graph-energy determinant. The difference from BR5 is precisely $-\mathcal{R} \log \det Z_N = o(kq)$. This identifies and bounds the entire change from observed coefficient coordinates to the energy-minimizing section equipped with its original norm.

3. A subexponential regularizer transfers the complete observed action

Let $M_{B,N} = \Lambda M L_{B,N}$, using the original attained target metric. For $z > 0$, ME8 and ME22 give the exact identity

$$\begin{aligned} \log \det(zI + M_{B,N}^\dagger M_{B,N}) &= \log \det(zI + H_N) - \log \det(zI + A_N) \\ &\quad - \ell_N(z) + g_N(z). \end{aligned} \quad (\text{BR7})$$

No coupling block is omitted. If the finite gap from ME29 is $A_N \succeq \alpha_N I$, diagonalization of A_N gives

$$0 \leq \log \det(zI + A_N) - \log \det A_N \leq m \log(1 + z/\alpha_N).$$

Together with ME24 this proves, for any four positive z_N ,

$$\begin{aligned} &\left| \mathcal{R} \log \det(z_N I + M_{B,N}^\dagger M_{B,N}) - \mathcal{R} \log \det(z_N I + H_N) \right| \\ &\leq \sum_N [|\log \det A_N| + m \log(1 + z_N/\alpha_N) + \mathcal{B}_m(z_N)]. \end{aligned} \quad (\text{BR8})$$

This is a finite bound. For $|\log z_N| = o(q)$, uniformly over the four endpoints, its right side is $o(kq)$: use ME34, $m = 8k - 16$, $\log^+(1/\alpha_N) = O_{h,A}(k \log(q+2))$, and $\log(1 + z/\alpha) \leq \log 2 + \log^+ z + \log^+(1/\alpha)$. ME25 bounds the last term. Thus the actual observed compression and the full canonical action have the same four-cutoff coefficient at scale kq on this regularizer range. This evaluated receiving result strengthens the earlier memory comparison by retaining the kernel block as well.

The positive-regularizer domain is necessary for this formula as written. ME26 gives its exact boundary correction: when $M_{B,N}$ has a kernel of dimension p_N , the leakage contains $-p_N \log z$. The original Schur energy quotient BR4 remains positive at zero. Its exact map and correction are BR4–6. The zero-regularizer memory bound alone does not delete this term.

4. All colliding filter directions enter the original quotient

For an admitted monic p_k of degree d_k , RF37–60 proves in the unchanged numerator frame

$$\mathcal{R} \log \det(U_{p_k}^* \Lambda^* Q_{B,N} \Lambda U_{p_k}) = C_\partial d_k q + o(d_k q), \quad d_k \log(q+2) = o(q). \quad (\text{BR9})$$

RF65–67 gives the exact original quotient π_{p_k} with kernel $\text{im } U_{p_k}$, including its low-polynomial right inverse. Its attained metric changes the original kernel determinant by at most $2 \min(m, d_k) E_{k,d_k,N}$ at each endpoint, which is $o(kq)$ after summing.

Thus the kernel coefficient remains attached to the same full original frame under the rational quotient. This extends the earlier inverse-power quotient receivers to every stated colliding compact pole family.

RM24–26 gives the complete observed residue matrix at every repeated slow eigenvalue. RM28–31 then transfers all d_k slow eigenvalues to the actual energy graph and proves the determinant return $-C_{\partial}d_kq + o(d_kq)$ for that cluster. The complete graph outside the cluster remains in the full matrix; BR6’s whole-graph zero coefficient concerns the canonical degree-one multiplication operator.

For the explicit supplied filter $p_k = [(y-13)(y-4)(y-18)(y-468)]^k$, RF61–64 derives the arithmetic witness, unchanged degree $4k$, physical-coordinate map and rank-one multiplication defect. BR9 evaluates its observed determinant as $4C_{\partial}kq + o(kq)$, with $5.415572 < 4C_{\partial} < 5.419004$. The unweighted full heat return divided by $4k$ tends to 2 at RF55’s original common time; the actual measured heat is eventually positive with logarithmic rate zero after division by $4k$, as proved in RF60. Its observation fractions remain explicit.

Proof sources and affected earlier results

The original complete kernel is MR1–23. BR1–6 supplies evaluated energy-action and graph corrections to that same determinant. The earlier memory construction is UH1–32 in the complete 014 reader; ME20 and BR7–8 evaluate its canonical receiving bound at scale kq , with the exact zero and positive regularizer domains stated above. The inverse-source receivers are AK5–12 and AK20; RF65–67 and BR9 extend them to polynomial filters. Complete source texts are retained with the edition.

The human Feshbach–Schur source used at RM18–26 is Geneviève Dusson, Israel Michael Sigal and Benjamin Stamm, *The Feshbach–Schur map and perturbation theory*, original author TeX, arXiv:2105.02058v1, theorem thm:isospF, reconstruction equation QP, and proposition prop:U-prop-SA. The finite proofs of every identity used here are written out in RM. The original Gamma and Meixner–Pollaczek human sources are cited in RF1–15 and retained with their original attribution in the source collection. The higher-cutoff coefficient uses the previously retained monic-profile and zero-law input at its stated analytic scope.

Part II

Canonical memory and the entire energy graph

Canonical memory at zero and the exact energy-graph determinant

Independent derivation, 21 September 2026. Equations **ME1-ME34** belong to this proof. The received claims addressed here are (1)-(9) and (47)-(49). Every finite assertion below is proved in its stated metric. The final native estimates use the retained original-source theorems identified in Section 1; their analytic hypotheses and eventual scope remain in force. The source profile is not represented as a new result of this derivation.

1. Original objects and the precise source receiver

Fix the original simple quartet, with $0 < \delta < 1/2$, $\gamma > 2$, $k \geq 9$, $k \equiv 1 \pmod{4}$, and

$$\begin{aligned} q &= (k+1)^2, & S &= k/2 + iy, \\ Q_k(y) &= \prod_{a,b=0}^k [y - (2b-k)\gamma + i(2a-k)\delta], \\ E_k &= \mathbb{C}[y]/(Q_k), & M_k[P] &= [yP]. \end{aligned} \tag{ME1}$$

The measure is exactly $d\mu_k = w_h^{*k}(y) dy$, with $w_h(y) = |(2\xi/h)(1/2 + iy)|^2/(2\pi)$. Its mass and the full divisor h are retained. For $q-1 \leq N \leq 2q$, the original metric G_N is the attained minimum over the whole fibre of the remainder map from polynomials of degree at most N . In particular its relation space contains every polynomial $Q_k P$ allowed by that degree.

The onto original observation $\Lambda : E_k \rightarrow B_k$, including all invariant columns, has

$$Q_{B,N} = (\Lambda G_N^{-1} \Lambda^*)^{-1}, \quad L_{B,N} = G_N^{-1} \Lambda^* Q_{B,N}, \quad P_{B,N} = L_{B,N} \Lambda, \quad K = \ker \Lambda. \tag{ME2}$$

Multiplication verifies $\Lambda L_{B,N} = I$, $L_{B,N}^* G_N L_{B,N} = Q_{B,N}$, and $P_{B,N}^2 = P_{B,N} = P_{B,N}^\dagger$. Thus $L_{B,N}$ is the exact isometry from the attained observed metric to $K^{\perp G_N}$, and $\Lambda = L_{B,N}^\dagger$. These maps are the ones used throughout.

The original rank is

$$m = \dim K = 8k - 16, \quad n = \dim B_k = q - m = k^2 - 6k + 17. \tag{ME3}$$

This is asserted on the proved five-orbit period locus, with the actual period, units and coefficient pivots retained. In the original frame, $Z = [M_A, S_k Z_k]$, $\ker \pi = \text{im } Z$, and $I_K = V^{-1} \pi^T$; the transpose is the complex-linear dual, not a metric adjoint. These formulas and the rank computation are MR1-8, original kernel source. The source's verified period domain $|u| \geq R_*$, and its warning that the size-one orbit has a different rank, are NG24, original Gaussian transfer. The finite matrix theorems below hold for the actual rank at any other period; ME3 and its asymptotic uses are confined to their proved period domain.

The native action has the exact original-metric decomposition

$$\begin{aligned}
M_k &= C_N + r_N \ell_N, & C_N^\dagger &= C_N, \\
r_N &= [p_{N+1}], & \ell_N(v) &= [y^N] L_N v, & \ell_N r_N &= 0, \\
\|C_N\| &\leq C_0 := B_h q, & \epsilon_N &= \|r_N \ell_N\|, & R_N &= \epsilon_N + C_0.
\end{aligned} \tag{ME4}$$

Here $L_N : E_k \rightarrow \mathcal{P}_N$ is the full remainder minimum section, distinct from $L_{B,N}$. The plus sign and parity statement are SD1-3. The actual full multiplication inequality is $\|yP\|_{\mu_k} \leq C_{\text{mult}}(h)(2L + \lfloor k/3 \rfloor) \|P\|_{\mu_k}$ for $\deg P \leq L$; see NG2-3. For example the original choice $B_h = 5 \max(1, C_{\text{mult}}(h))$ covers $N \leq 2q$. Taking $L = N$ in that inequality bounds the complete degree-raising multiplication before compression; the degree $N + 1$ column is included.

The retained H2/H5 theorem explicitly includes $N = 2q$ and states, uniformly in this cutoff window,

$$\log \epsilon_N = q\psi(s_N) + O_h(k \log^2(q+2)), \quad s_N = (N+1-q)/q. \tag{ME5}$$

Its continuous decreasing profile has $\psi(0) = \log(4/\pi)$ and $\psi(1) > 0$. The exact source identity and its inherited EIQ input are H1-9, retained profile. In particular $s_{2q} = 1 + 1/q$ is covered. This is an eventual analytic theorem, not an effective numerical threshold in k .

All roots in ME1 are nonzero, since both odd integers $2a - k, 2b - k$ are nonzero. Thus M_k is invertible at every stated finite cutoff. Its determinant is metric independent and exactly

$$|\det M_k|^2 = |Q_k(0)|^2 = \prod_{a,b=0}^k [(2a-k)^2 \delta^2 + (2b-k)^2 \gamma^2]. \tag{ME6}$$

2. The full memory, including zero regularizer

For this section let E be any q -dimensional positive Hilbert space, $K \subset E$, $0 < m = \dim K < q$, and let $M : E \rightarrow E$ be invertible. Use the metric orthogonal splitting $E = K \oplus K^\perp$ and its isometric inclusions. Write

$$\begin{aligned}
M &= \begin{pmatrix} M_K & D \\ C & M_B \end{pmatrix}, & H &= M^\dagger M = \begin{pmatrix} A & B_H \\ B_H^\dagger & D_H \end{pmatrix}, \\
A &= M_K^\dagger M_K + C^\dagger C, & D_H &= D^\dagger D + M_B^\dagger M_B.
\end{aligned} \tag{ME7}$$

In particular H, A, D_H are positive definite. Define, now for every $z \geq 0$,

$$\begin{aligned}
F(z) &= zI + D_H - B_H^\dagger(zI + A)^{-1} B_H, \\
g(z) &= \log \det(zI + A) + \log \det(zI + D_H) - \log \det(zI + H).
\end{aligned} \tag{ME8}$$

Block Gaussian elimination, whose triangular factors have determinant one, proves $\det(zI + H) = \det(zI + A) \det F(z)$. Completing the positive quadratic form shows $F(z) > 0$, even at $z = 0$. Thus all logarithms in ME8 are defined there, and g is continuous at zero.

Put

$$\mathcal{Q}(t) = \text{Tr} e^{-tH} - \text{Tr} e^{-tA} - \text{Tr} e^{-tD_H}. \tag{ME9}$$

For completeness, choose orthonormal eigenbases of A and D_H and concatenate them. If e belongs to this basis, scalar Jensen applied to the spectral decomposition of H gives $\langle e, e^{-tH} e \rangle \geq e^{-t\langle e, He \rangle}$, because the weights $|\langle e, v_j \rangle|^2$ are nonnegative and sum

to one, and $x \mapsto e^{-tx}$ is convex. Summing proves $\mathcal{Q}(t) \geq 0$. The Taylor series gives $\mathcal{Q}(t) = t^2 \|B_H\|_{\text{HS}}^2 + O(t^3)$: the constant and linear terms cancel, and $\text{Tr } H^2 - \text{Tr } A^2 - \text{Tr } D_H^2 = 2\|B_H\|_{\text{HS}}^2$. At infinity all three traces decay exponentially at each fixed finite matrix. Consequently

$$\boxed{g(z) = \int_0^\infty e^{-zt} \mathcal{Q}(t) \frac{dt}{t}, \quad z \geq 0; \quad 0 \leq g(z) \leq g(0).} \quad (\text{ME10})$$

To verify the integral identity without interchanging divergent pieces, pair the q eigenvalues of $A \oplus D_H$ with the q eigenvalues of H . Each paired scalar integral is $\int_0^\infty (e^{-at} - e^{-bt}) dt/t = \log(b/a)$ for $a, b > 0$: differentiate with respect to b , integrate e^{-bt} , and use equality at $a = b$. Sum finitely many pairs. The estimates just proved justify the combined integral at zero. They also justify every derivative, including its one-sided value at zero: $(-1)^j g^{(j)}(z) = \int_0^\infty t^{j-1} e^{-zt} \mathcal{Q}(t) dt \geq 0$ for $j \geq 1$. Thus the complete monotonicity extends to this finite zero endpoint. No lower spectral gap uniform in k has been assumed.

3. Sharp restriction determinants with one exceptional singular value

Assume the descending singular values of M obey

$$s_1(M) \leq R, \quad s_j(M) \leq C_0 \ (j \geq 2), \quad R \geq C_0 > 0. \quad (\text{ME11})$$

The native ME4 implies this: on $\ker \ell_N$, a codimension-one space, $M = C_N$ and hence $\|M\| \leq C_0$ there. The variational characterization of the second singular value gives $s_2 \leq C_0$; triangle inequality gives $s_1 \leq R_N$. The variational characterization follows directly by intersecting any subspace of the required dimension with the span of the ordered eigenvectors of $M^\dagger M$.

For every isometry $J : \mathbb{C}^d \rightarrow E$, $1 \leq d \leq q$, restriction and the same variational argument give $\lambda_j(J^\dagger H J) \leq s_j(M)^2$. Therefore

$$\boxed{\det(J^\dagger H J) \leq R^2 C_0^{2(d-1)}}. \quad (\text{ME12})$$

This is sharp, by restricting $\text{diag}(R^2, C_0^2, \dots, C_0^2)$ to a subspace containing its first coordinate. In particular the received two separate bounds for $\det A$ and $\det D_H$ follow.

They can be sharpened jointly. Let v be a unit top eigenvector of H , and put $t = \|P_K v\|^2$. Spectral decomposition gives $H \preceq C_0^2 I + (R^2 - C_0^2) v v^\dagger$. Compression, the rank-one determinant formula and positive determinant monotonicity imply

$$\begin{aligned} \det A \det D_H &\leq C_0^{2(q-2)} [C_0^2 + (R^2 - C_0^2)t] [C_0^2 + (R^2 - C_0^2)(1-t)] \\ &= C_0^{2(q-2)} [R^2 C_0^2 + (R^2 - C_0^2)^2 t(1-t)] \\ &\leq C_0^{2(q-2)} \frac{(R^2 + C_0^2)^2}{4}. \end{aligned} \quad (\text{ME13})$$

Positive determinant monotonicity used here follows by congruence: if $0 < X \preceq Y$, all eigenvalues of $X^{-1/2} Y X^{-1/2}$ are at least one. The last bound in ME13 is sharp for every nontrivial splitting: take $H = C_0^2 I + (R^2 - C_0^2) v v^\dagger$, with $\|P_K v\|^2 = 1/2$. Its positive square root supplies an invertible M attaining equality. This proves optimality of this universal numerator bound, not merely of the two separate factors.

Combining ME10 and ME13 proves the uniformly valid improvement of the supplied (3):

$$0 \leq g(z) \leq g(0) \leq 2 \log \frac{R^2 + C_0^2}{2} + 2(q-2) \log C_0 - 2 \log |\det M|, \quad z \geq 0. \quad (\text{ME14})$$

Since $(R^2 + C_0^2)/2 \leq R^2$, the received $4 \log R$ upper bound is valid. More precisely the same argument for $zI + H$ gives the finite regularized estimate

$$g(z) \leq (q-2) \log(z + C_0^2) + 2 \log \left(z + \frac{R^2 + C_0^2}{2} \right) - \log \det(zI + H). \quad (\text{ME15})$$

The determinant denominator is the complete action determinant, not the determinant of its bounded compression. If one additionally knows $\sigma_1(M^{-1}) \leq L$, $\sigma_j(M^{-1}) \leq D_0$ for $j \geq 2$, one obtains the useful independent bound

$$g(0) = \log \det A + \log \det((H^{-1})_{KK}) \leq 2 \log(RL) + 2(m-1) \log(C_0 D_0). \quad (\text{ME16})$$

Indeed Schur elimination against D_H proves $(H^{-1})_{KK} = (A - B_H D_H^{-1} B_H^\dagger)^{-1}$, giving the equality; apply ME12 to H and H^{-1} . The minimum of ME14-16 and the local bound below may always be used.

4. Native coefficient and the uniform zero endpoint

Define

$$I_\square = \log(\delta^2 + \gamma^2) - 3 + \frac{\delta}{\gamma} \arctan \frac{\gamma}{\delta} + \frac{\gamma}{\delta} \arctan \frac{\delta}{\gamma}. \quad (\text{ME17})$$

It is $\int_0^1 \int_0^1 \log(\delta^2 x^2 + \gamma^2 y^2) dx dy$. For example first integrate in x , using the antiderivative $x \log(\delta^2 x^2 + \gamma^2 y^2) - 2x + (2\gamma y/\delta) \arctan(\delta x/(\gamma y))$; a second integration gives exactly ME17, including the constant -3 .

Here is the finite lattice estimate with every root retained. Put $n_* = k + 1$, and use midpoint coordinates $x_a = (2a - k)/n_*$ on $[-1, 1]$. Then ME6 is $\log |\det M_k|^2 = 2q \log n_* + \sum_{a,b} f(x_a, x_b)$, $f(x, y) = \log(\delta^2 x^2 + \gamma^2 y^2)$. The four central cells have centre-minus-cell-average difference $\log(\delta^2 + \gamma^2) - \log 4 - I_\square$. On shell $j \geq 1$, whose maximum centre-coordinate magnitude is $(2j + 1)/n_*$, there are $8j + 4 \leq 12j$ cells. Each point in them has radius at least $2j/n_*$, and $\|\nabla f\| \leq 2\gamma^2/(\delta^2 \sqrt{x^2 + y^2})$. The distance from a cell centre is at most $\sqrt{2}/n_*$. Each error is consequently at most $\sqrt{2}\gamma^2/(\delta^2 j)$. Summing at most $n_*/2$ shells proves

$$\begin{aligned} |\log |\det M_k|^2 - 2q \log k - q I_\square| &\leq E_{\text{root}}(k), \\ E_{\text{root}}(k) &:= 2q \log \frac{k+1}{k} + \frac{6\sqrt{2}\gamma^2}{\delta^2} (k+1) \\ &\quad + 4 |\log(\delta^2 + \gamma^2) - \log 4 - I_\square|. \end{aligned} \quad (\text{ME18})$$

This reproduces HAR33-35 and proves the claimed $O_{\delta,\gamma}(k)$ error independently from the displayed cells. In particular no logarithmic cell error has been lost at the four innermost roots.

ME5 and continuity with $\psi(1) > 0$ imply that $\epsilon_N/C_0 \rightarrow \infty$ uniformly in the window. Indeed the profile has a fixed positive lower bound on $[0, 1 + 1/q]$ for all sufficiently large q , while $k \log^2(q+2) = o(q)$. Thus $\log R_N = \log \epsilon_N + \log(1 + C_0/\epsilon_N) = q\psi(s_N) + O_h(k \log^2(q+2))$. Substituting the weaker form of ME14 and ME18, and expanding $2(q-2) \log(B_h q)$, gives

$$g_N(z) \leq 2q \log(q/k) + q[4\psi(s_N) + 2 \log B_h - I_\square] + O_h(k \log^2(q+2) + \log(q+2)), \quad z \geq 0. \quad (\text{ME19})$$

The exact finite alternative, requiring no asymptotic threshold, is ME14 with R_N , $B_h q$, and the root product ME6, or with the explicitly bounded error ME18. The coefficient $2q \log(q/k)$ in the received (4) is correct. In particular inserting an additional $q \log q$ into $\log \epsilon_N$, or replacing $\log |\det M|$ by $\log |\det M|^2$ in (3), would be an error. The retained source H5 concerns $\log \epsilon_N^2 = 2q\psi + O_h(k \log^2 q)$.

Since $q = (k+1)^2$, $\log(q/k) = O(\log k) = o(k)$, ME19 proves

$$\sup_{q-1 \leq N \leq 2q} \sup_{z \geq 0} g_N(z) = O_h(q \log(q/k)) = o_h(kq), \quad \int_0^\infty \mathcal{Q}_N(t) \frac{dt}{t} = g_N(0). \quad (\text{ME20})$$

This uniform assertion uses positivity and the exact zero determinant; it does not require a uniform decay rate for the heat integrand. For the original signs $+, +, -, -$ at $q-1, q, 2q-1, 2q$, the absolute value of the signed sum at any four nonnegative endpoint-dependent regularizers is bounded by four times the supremum in ME20. Its quotient by kq tends to zero. This proves the received (1), (5) and (6), including independent regularizers equal to zero.

5. Local leakage and its exact difference from memory

For $z > 0$ put

$$\ell(z) = \log \det(zI + D_H) - \log \det(zI + M_B^\dagger M_B). \quad (\text{ME21})$$

Since $D_H = M_B^\dagger M_B + D^\dagger D$, $\ell \geq 0$. ME8 yields exactly

$$\log \det F(z) - \log \det(zI + M_B^\dagger M_B) = \ell(z) - g(z). \quad (\text{ME22})$$

In original observed coordinates M_B is precisely $\Lambda M L_{B,N}$, with the $Q_{B,N}$ adjoint. Thus ME22 is already the requested original receiving map, with both coupling blocks retained.

Let $r_D = \text{rank } D$, $r_H = \text{rank } B_H$, and define $\mathcal{B}_0(z) = 0$,

$$\mathcal{B}_r(z) = \log(1 + R^2/z) + (r-1) \log(1 + C_0^2/z), \quad r \geq 1. \quad (\text{ME23})$$

Then the rank-sensitive strengthening of the supplied (8) is

$$0 \leq \ell(z) \leq \mathcal{B}_{r_D}(z), \quad 0 \leq g(z) \leq \mathcal{B}_{r_H}(z), \quad |\ell(z) - g(z)| \leq \mathcal{B}_{\max(r_D, r_H)}(z) \leq \mathcal{B}_m(z). \quad (\text{ME24})$$

Here $r_D, r_H \leq \min(m, n)$. To prove the first bound, apply the determinant identity to $zI + M_B^\dagger M_B$ and $D^\dagger D$; the nonzero eigenvalues of the resulting relative matrix are bounded by $z^{-1} s_j(D)^2$. Restricting and projecting M cannot increase its ordered singular values: $(PMJ)^\dagger PMJ \preceq J^\dagger HJ$, followed by the variational argument of ME12. Thus $s_1(D) \leq R$, $s_j(D) \leq C_0$ for $j \geq 2$, and only r_D factors occur.

For memory, eliminate the other diagonal block. Set $T_z = B_H(zI + D_H)^{-1} B_H^\dagger$ and $F_K = zI + A - T_z$. Positivity of H implies $B_H D_H^{-1} B_H^\dagger \preceq A$, so $0 \preceq T_z \preceq A$, $F_K \succeq zI$, and $\text{rank } T_z = r_H$. Now $g = \log \det(I + F_K^{-1/2} T_z F_K^{-1/2})$. Its nonzero eigenvalues are those of $T_z^{1/2} F_K^{-1} T_z^{1/2}$, bounded by the ordered eigenvalues of T_z/z , hence by those of

A/z . ME12 supplies ME24. The final difference bound uses that both numbers are nonnegative; it does not require adding their two upper bounds.

For native matrices, $\log(1+R_N^2) = O_h(q)$, $\log(1+C_0^2) = O_h(\log(q+2))$, and $m = 8k-16$. The elementary inequality $\log(1+x/z) \leq \log(1+x) + \log^+(1/z)$, $x \geq 0$, gives

$$|\ell_N(z) - g_N(z)| \leq O_h(q + k \log(q+2)) + (8k-16) \log^+(1/z). \quad (\text{ME25})$$

Therefore the supplied (9) is correct whenever $z_k > 0$ and $\log^+(1/z_k) = o(q)$. A slightly stronger exact statement replaces $8k-16$ in its variable-regularizer cost by $\max(r_D, r_H)$. Nothing in ME20 permits dropping the positive-regularizer guard from the separately observed action.

Specifically, if $p = \dim \ker M_B$, diagonalization of $M_B^\dagger M_B$ gives

$$\ell(z) = -p \log z + \log \det D_H - \log \text{pdet}(M_B^\dagger M_B) + o(1), \quad z \downarrow 0. \quad (\text{ME26})$$

The empty positive spectral product is one. The numerator is continuous because $D_H > 0$. This proves exactly when leakage diverges. For the rational fixture $M = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $K = \mathbb{C}e_1$, one has $H = I$, $g \equiv 0$, $M_B = 0$, and $\ell(z) = \log(1+1/z)$. Thus bounded memory alone cannot bound the zero-regularizer comparison with the independently compressed action.

6. Finite lower kernel gap and the native guard

Let the smallest eigenvalue of H be simple, equal to $h > 0$, with unit eigenvector w , and suppose all remaining eigenvalues are at least $\gamma_* > h$. Put $\theta = \|P_B w\|^2$. For a unit $x \in K$, $|\langle w, x \rangle|^2 \leq \|P_K w\|^2 = 1 - \theta$. Spectral decomposition then gives

$$A \succeq [h(1-\theta) + \gamma_* \theta] I_K \succeq \gamma_* \theta I_K. \quad (\text{ME27})$$

This proves the supplied lower bound from the actual observed fraction of the actual singular line. To evaluate it natively, retain the exact inverse identity

$$\begin{aligned} M_k^{-1} &= B_N - \frac{u \varphi_N}{Q_k(0)}, & B_N &= J_N \mathcal{D} L_N, & \mathcal{D} p &= (p - p(0))/y, \\ u &= J_N \frac{Q_k - Q_k(0)}{y} = -Q_k(0)[y^{-1}], & \varphi_N(v) &= (L_N v)(0). \end{aligned} \quad (\text{ME28})$$

It follows by multiplying by y modulo the full polynomial. The finite original bound is $\|B_N\| \leq d_N := \sqrt{u_k/\ell_{k,N}} \sqrt{n_N(n_N+1)(1+\sqrt{N+1})}$, $n_N = \lceil (N+2)/2 \rceil$, with $\log d_N = O_h(k + \log q)$; see S11-16. These are the exact full-mass NG2 comparison constants.

Put $E_N = \|u\| \|\varphi_N\| / |Q_k(0)|$, $\eta_N = d_N / (E_N - d_N)$, and $\theta_u = \|P_B u\|^2 / \|u\|^2$, an exact quotient-Gram scalar. On the explicit finite guards $E_N > 2d_N$ and $\theta_u > \eta_N$, a usable lower bound is

$$\alpha_N = d_N^{-2}(\theta_u - \eta_N) > 0, \quad A \succeq \alpha_N I. \quad (\text{ME29})$$

Proof: on $\ker \varphi_N$, $M^{-1} = B_N$, so $\sigma_2(M^{-1}) \leq d_N$; triangle inequality gives $\sigma_1(M^{-1}) \geq E_N - d_N > d_N$. The smallest singular value of M is therefore simple, and the rest of the eigenvalues of H are at least d_N^{-2} . A unit left top singular vector w of M^{-1} obeys $\|(I - P_u)w\| \leq \eta_N$, by applying $I - P_u$ to ME28. The difference $P_w - P_u$ has eigenvalues $+s, -s$, $s \leq \eta_N$; in an orthonormal pair this follows by expanding the two rank-one projections. For any orthogonal projection P_B , its trace against this difference is between $-s$ and s , since the two diagonal projection entries lie in $[0, 1]$.

Thus $\theta \geq \theta_u - \eta_N$, and ME27 proves ME29. This slightly improves the safe trace-norm error $2\eta_N$.

The original conductor comparison for the inverse-power rank $r = 1$ proves $\theta_u \geq \vartheta_{1,N} > 0$ and $\log(1/\vartheta_{1,N}) = O_{h,A}(k \log(q+2))$. Its fully specified domain is IVO24-26: $q_- = (k-7)^2 \geq 16$, $k\sqrt{\delta^2 + \gamma^2} \leq 2^{-14}q_-$, $n_- + 2 \leq 2q_-$, and $\mathcal{L}_*\mathcal{R}_*T_N^{\text{lb}} \geq 2\mathcal{A}$, for an actual nonzero conductor coefficient, with exactly the constants defined there. These finite source conditions are not inferred from the small fixture $k = 9$. IVO26 proves they hold eventually for fixed original data and $r = 1$. The received inverse-observation asymptotic is used only through this proved finite angle estimate.

The retained original smallest-singular theorem S4-6 gives $\log \|M_k^{-1}\| = q \log(q/k) + O_h(q)$ throughout this cutoff window. Because $|E_N - \|M_k^{-1}\|| \leq d_N$, one has $\eta_N = \exp[-q \log(q/k) + O_h(q)]$. It is eventually smaller than $\vartheta_{1,N}/2$, uniformly in the four cutoffs. Thus the fully source-evaluated finite guard $E_N > 2d_N$, $\eta_N < \vartheta_{1,N}/2$ implies $\alpha_N \geq \vartheta_{1,N}/(2d_N^2)$, and $\log^+(1/\alpha_N) = O_{h,A}(k \log(q+2))$. All eventual state-ments retain fixed h, δ, γ and the original period; no arbitrary high-degree filter is substituted for M_k .

7. Exact energy graph, its determinant cost, and coefficient coordinates

Return first to any invertible finite M . Write $B = B_H$, and suppose a finite $\alpha > 0$ with $A \succeq \alpha I$ has been evaluated, for example by ME29. For each $b \in K^\perp$, completion of the square gives

$$\langle (k, b), H(k, b) \rangle = \langle k + A^{-1}Bb, A(k + A^{-1}Bb) \rangle + \langle b, Sb \rangle, \quad S = D_H - B^\dagger A^{-1}B > 0. \quad (\text{ME30})$$

Hence the exact energy minimum section is $L_{\text{en}}b = (-A^{-1}Bb, b)$. Its original norm Gram is $Z = I + B^\dagger A^{-2}B$, its energy Gram is S , and $J_{\text{en}} = L_{\text{en}}Z^{-1/2}$ is an isometry. Direct multiplication gives $J_{\text{en}}^\dagger H J_{\text{en}} = Z^{-1/2}S Z^{-1/2}$.

Let $J_K : K \rightarrow E$ denote the isometric inclusion, not the unadjusted coefficient frame I_K . If that frame is used, $J_K = I_K(I_K^*GI_K)^{-1/2}$. Since $A = J_K^\dagger H J_K$, define the explicitly specified isometry

$$V = H^{1/2}J_K A^{-1/2} : K \longrightarrow E.$$

Indeed $V^\dagger V = I$. Squaring the complete block matrix gives $J_K^\dagger H^2 J_K = A^2 + BB^\dagger$. The rectangular determinant identity $\det(I + UV) = \det(I + VU)$, proved by eliminating either diagonal block in $\begin{pmatrix} I & U \\ -V & I \end{pmatrix}$, therefore yields

$$\det Z = \frac{\det(J_K^\dagger H^2 J_K)}{(\det A)^2}, \quad \det A \det Z = \det(V^\dagger H V). \quad (\text{ME31})$$

This identity keeps the exact map relating the energy-weighted kernel subspace $H^{1/2}K$ to the original kernel. It is not a replacement of the physical metric. Applying ME12 both to J_K and V , and using $Z \succeq I$, gives, with $D_* := 2 \log R + 2(m-1) \log C_0$,

$$m \log \alpha \leq \log \det A \leq D_*, \quad 0 \leq \log \det Z \leq D_* - \log \det A \leq D_* - m \log \alpha. \quad (\text{ME32})$$

This proves (47) and improves (48). The useful combined estimate is $|\log \det A| + \log \det Z \leq D_* + 2m \log^+(1/\alpha)$: if $a = \log \det A \geq 0$, its left side is at most D_* ; if $a < 0$, it is at most $D_* - 2a$. This proof works for every $m \geq 1$, including $m = 1$, without a negative multiplicity $m - 2$.

For comparison, the received rank-two estimate is also valid for the native $m \geq 2$, but a complete finite form must account for the domain and codomain ranks. Namely $H - C_N^2 = Cr\ell + \ell^\dagger r^\dagger C + \|r\|^2 \ell^\dagger \ell$ has range in $\text{span}(Cr, \ell^\dagger)$, so its rank is at most two. Consequently $A^{-1}B$ differs by rank at most two from $A^{-1}P_K C_N^2|_{K^\perp}$, whose norm is at most C_0^2/α . Restriction to the kernel of that difference proves that every singular value after the second is at most C_0^2/α . All are at most R^2/α . Writing $r = \min(m, n)$, $t = \min(2, r)$, one obtains $\log \det Z \leq t \log(1 + R^4/\alpha^2) + (r - t) \log(1 + C_0^4/\alpha^2)$. For actual $m = 8k - 16$, this implies the supplied (48) by increasing the count to m . The received expression is thus a valid weaker bound on its native domain, with a multiplicity repair required only if it is claimed for arbitrary small dimensions.

Finally express the result in the original observed coefficient coordinates. Put $G = G_N$, $Q = Q_{B,N}$, and choose the square isometry $U = [J_K, L_{B,N}Q^{-1/2}]$, so $U^*GU = I$ and $\Lambda U = [0, Q^{-1/2}]$. Let $H_U = U^*M^*GMU$. Then $(M^*GM)^{-1} = UH_U^{-1}U^*$, and the observed block of H_U^{-1} is S^{-1} by elimination. Therefore the exact energy quotient and its graph compression satisfy

$$\begin{aligned} \mathcal{E}_B &= [\Lambda(M^*GM)^{-1}\Lambda^*]^{-1} = Q^{1/2}SQ^{1/2}, \\ \det(Q^{-1/2}\mathcal{E}_BQ^{-1/2}) &= \det S = \frac{|\det M|^2}{\det A}, \\ \det(J_{\text{en}}^\dagger H J_{\text{en}}) &= \frac{|\det M|^2}{\det A \det Z} = \frac{|\det M|^2}{\det(V^\dagger H V)}. \end{aligned} \tag{ME33}$$

In arbitrary observed coordinates the energy-minimizing lift is exactly $L_{B,N} - J_K A^{-1}BQ^{1/2}$; its projection under Λ is the identity, and its norm Gram is $Q^{1/2}ZQ^{1/2}$. Thus the coefficient, energy, and isometric graph objects in ME33 are related by stated maps, with no unidentified determinant factor.

For the original canonical M_k , ME4-5 give $D_* = O_h(q + k \log(q + 2))$, while ME29 gives $\log^+(1/\alpha_N) = O_{h,A}(k \log(q + 2))$ eventually on the stated finite guards. With the actual $m = 8k - 16$, ME32 proves

$$|\log \det A_N| + \log \det Z_N = O_{h,A}(q \log(q + 2)) = o_{h,A}(kq). \tag{ME34}$$

The same estimate applies to $|\log \det(V_N^\dagger H_N V_N)|$ by ME31. The numerator $|\det M_k|^2$ in ME33 is exactly independent of N ; it cancels under the original four signs. Consequently both four-cutoff logarithmic determinant returns in ME33, divided by kq , tend to zero. This proves (49) and the energy-graph consequence on the original canonical action. It makes no assertion that the same determinant cost holds for arbitrary rational filters or arbitrary high powers of M_k .

8. Verification scope

The accompanying `check_memory_energy.py` passes **208 exact rational checks** and **684 numerical fixture checks**. It checks the attained coefficient-to-isometry map, the sharp common-top-direction determinant, the leakage divergence fixture, and noncommuting energy graphs. The numerical section checks the zero-endpoint heat integral and inequalities on declared finite synthetic matrices; its largest graph-identity discrepancy is less than 6.16×10^{-14} . These are independent algebraic and numerical fixtures, not computed native xi moments, period evaluations, or a replacement for the proof. `MEMORY_ENERGY_CHECK_RESULTS.json` retains the individual checks. The complete source identity and reading coverage are recorded separately in `MEMORY_SOURCE_USE_PRIVATE.json`, with eight local source hashes matched to their retained full-byte public receipts.

An exact two-dimensional geometric fixture is $G = I$, $K = \mathbb{R}e_1$, $M = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$, $H = \begin{pmatrix} 5 & 4 \\ 4 & 5 \end{pmatrix}$. It attains ME13: $R = 3$, $C_0 = 1$, $t = 1/2$, $\det A \det D_H = 25$, $g(0) = \log(25/9)$. Its actual energy section is $b \mapsto (-4b/5, b)$, $Z = 41/25$, $S = 9/5$, and $V(1) = (2, 1)/\sqrt{5}$. Thus $\det A \det Z = 41/5 = V^T H V$, while the isometric energy-graph compression is $45/41$. These coordinates show the two distinct subspaces and their exact maps in ME30-33.

Part III

Colliding polynomial filters in the original source

Colliding rational filters in the complete original metric

21 September 2026. Independent derivation of received equations (10)–(32) and (50)–(58). The exact received source and its hash are retained in the source ledger. Every norm below uses the original source or its stated attained minimum. The full observed fibre, all conductor coefficients, and every filter multiplicity are retained.

The collision-uniform theorem is valid. This proof supplies explicit finite constants and guards for its contour, lower relation degree, numerator absorption, surjective remainder map, and separated singular cluster. The evaluated coefficient uses the established original monic profile and zero law at their stated scope; the finite inequalities do not require those asymptotic inputs. The exact checks accompanying this proof test algebra and matrix identities, not unknown native moments.

1. Objects, coordinates and source inputs

Fix the original simple-quartet and period domain, with

$$0 < \delta < \frac{1}{2}, \quad \gamma > 2, \quad k \geq 9, \quad k \equiv 1 \pmod{4}, \quad q = (k+1)^2, \quad S = k/2 + iy, \\ Q = Q_k(y) = \prod_{a,b=0}^k [y - (2b-k)\gamma + i(2a-k)\delta], \quad E = \mathbb{C}[y]/(Q), \quad M[f] = [yf]. \quad (\text{RF1})$$

Write $D_0 = \sqrt{\delta^2 + \gamma^2}$, $R_k = kD_0$. Keep $d\mu_k = w_h^{*k}(y) dy$ and

$$d\sigma(y) = |\Gamma(1/4 + iy/2)|^2 dy / (2\pi), \quad M_\sigma = \sqrt{2\pi}.$$

The source degree is $q - 1 \leq N \leq 2q$. Its quotient map and full minimum section are $J_N : \mathcal{P}_N \rightarrow E$ and $L_N : E \rightarrow \mathcal{P}_N$. The attained metric is G_N , so J_N is contractive and L_N is an isometry. Original NG2 gives

$$\ell_{k,N} \|f\|_\sigma^2 \leq \|f\|_{\mu_k}^2 \leq u_k \|f\|_\sigma^2, \quad \log(u_k / \ell_{k,N}) = O_h(k + \log(N+1)). \quad (\text{RF2})$$

The unchanged onto observation is $\Lambda : E \rightarrow B$, with

$$Q_{B,N} = (\Lambda G_N^{-1} \Lambda^*)^{-1}, \quad L_{B,N} = G_N^{-1} \Lambda^* Q_{B,N}, \quad P_B = L_{B,N} \Lambda. \quad (\text{RF3})$$

Thus P_B is the original orthogonal projection, with kernel K , and

$$\|P_B x\|_{G_N} = \inf_{\deg f \leq N, \Lambda[f] = \Lambda x} \|f\|_{\mu_k}. \quad (\text{RF4})$$

The finite inputs used below are the proved IVO3-12 and IVO21-23, from the pinned complete IVO proof. They retain the original-author Meixner-Pollaczek generating function and recurrence of T. H. Koornwinder, R. Wong, R. Koekoek and R. F. Swarttouw, with W. P. Reinhardt, DLMF 18.23.7 and 18.22.8. R. A. Askey and R. Roy's Gamma product 5.8.3 supplies the retained Gamma envelope. Their original TeX versions were retained and read in the IVO source work; the source-use ledger identifies the exact local providers reread here.

Let

$$\mathcal{L} = \{(2a+1)\gamma - i(2b+1)\delta : a, b \in \mathbb{Z}\}, \quad \mathcal{P} \in \mathbb{C} \setminus \mathcal{L}, \quad R = \max_{\zeta \in \mathcal{P}} |\zeta|.$$

Assume $\mathcal{P} \neq \emptyset$. Every filter is monic,

$$p(y) = \prod_{\nu=1}^d (y - \lambda_\nu), \quad \lambda_\nu \in \mathcal{P}, \quad 1 \leq d \leq q,$$

with repetitions retained. Since the original roots belong to $\mathcal{L}$, p is a unit modulo Q . Define in the original numerator coefficient frame

$$U_p a = [a/p], \quad \deg a < d, \quad H_{p,N} = U_p^* G_N U_p, \quad H_{p,N}^B = U_p^* \Lambda^* Q_{B,N} \Lambda U_p. \quad (\text{RF5})$$

Multiplying $U_p a = 0$ by the unit p gives $[a] = 0$; its degree $< d \leq q$ proves $a = 0$. Thus $H_{p,N} > 0$ before any observed estimate.

2. A fixed contour and a complete collision-uniform remainder

The positive number $\eta = \text{dist}(\mathcal{P}, \mathcal{L})$ exists because the first set is compact and the second closed. Choose $a_\Gamma \in (R+1, R+2)$ different from every lattice modulus in that bounded interval. Then $b_\Gamma = \text{dist}(\{|z| = a_\Gamma\}, \mathcal{L}) > 0$. Choose

$$0 < \varepsilon_\Gamma < \min(\eta/4, \delta/4, b_\Gamma/3, 1/4).$$

Let Ω be the disk $|z| < a_\Gamma$ with the closed radius- ε_Γ disks about every lattice point inside it removed. These disks are disjoint, lie strictly inside the outer circle, and miss $\mathcal{P}$. Put $\Gamma = \partial\Omega$, with the outer circle positive and all hole circles negative. Then

$$\rho_\Gamma = \text{dist}(\Gamma, \mathcal{P}) > 0, \quad b = \text{dist}(\Gamma, \mathcal{L}) > 0, \quad |\Gamma| = 2\pi a_\Gamma + 2\pi \varepsilon_\Gamma \#(\mathcal{L} \cap \{|z| < a_\Gamma\}). \quad (\text{RF6})$$

This construction is independent of k, d, p . Every pole is enclosed with winding number one and every lattice point with winding number zero.

For f holomorphic on a neighbourhood of $\bar{\Omega}$, set

$$\mathcal{R}_p f(z) = \frac{1}{2\pi i} \int_\Gamma \frac{f(w)}{p(w)} \frac{p(w) - p(z)}{w - z} dw. \quad (\text{RF7})$$

The divided polynomial in z has degree $< d$. Cauchy's formula writes the expression near each pole as $f(z) - p(z)g(z)$ with g holomorphic there. Hence all derivatives through the pole's full multiplicity agree. This proves the complete Hermite remainder, including coincident poles. It also proves uniqueness: the difference of two such degree- $< d$ polynomials is divisible by p .

For every coefficient of $(p(w) - p(z))/(w - z)$, the elementary symmetric coefficient bounds give

$$\left| [z^j] \frac{p(w) - p(z)}{w - z} \right| \leq (R + \max(1, a_\Gamma))^d.$$

Indeed $[y^t]p \leq \binom{d}{t} R^{d-t}$, and the relevant finite geometric sum is bounded by the binomial expansion with $\max(1, |w|)$. Since $|p(w)| \geq \rho_\Gamma^d$,

$$\| \text{coeff } \mathcal{R}_p f \|_2 \leq C_d \sup_\Gamma |f|, \quad C_d = \frac{|\Gamma|}{2\pi} \sqrt{d} \left(\frac{R + \max(1, a_\Gamma)}{\rho_\Gamma} \right)^d. \quad (\text{RF8})$$

There is no inverse distance between poles.

For later use write $H_{\text{odd}}(t) = \sum_{1 \leq j \leq t, j \text{ odd}} j^{-1}$. The exact negation-pair shell count gives

$$\sum_{\omega/\pm, \omega \in Q_k} \frac{1}{|\omega|^2} \leq \frac{2}{\delta^2} H_{\text{odd}}(k). \quad (\text{RF9})$$

Consequently, for $|z| \leq r$,

$$|Q_k(z)/Q_k(0)| \leq M_k^+(r) := \exp[2r^2 H_{\text{odd}}(k)/\delta^2]. \quad (\text{RF10})$$

For the reciprocal on Γ , take one representative from every infinite-lattice negation pair of modulus at most $2a_\Gamma$, and put

$$C_\Gamma^- = \prod_{\substack{\omega/\pm \in \mathcal{L} \\ |\omega| \leq 2a_\Gamma}} \max(1, |\omega|^2/b^2), \quad M_k^- = C_\Gamma^- \exp[8a_\Gamma^2 H_{\text{odd}}(k)/(3\delta^2)]. \quad (\text{RF11})$$

Each retained small factor is bounded using $|(z - \omega)(z + \omega)| \geq b^2$. For all remaining factors, $|z^2/\omega^2| \leq 1/4$ and $-\log|1 - z^2/\omega^2| \leq (4/3)|z^2/\omega^2|$. Thus $\sup_\Gamma |Q_k(0)/Q_k| \leq M_k^-$. Both logarithmic costs are $O_{\mathcal{P},h}(\log(k+2))$, with every small lattice factor present.

3. Evaluation and division in the full Gamma source

Retain $b_L = \sqrt{2L+1} \sum_{j=1}^L 1/j$, $b_0 = 0$. IVO5 proves, for every complex z and polynomial of degree at most L ,

$$(2b_{L+1})^{-1} \|f\|_\sigma \leq \|(y - z)f\|_\sigma \leq [2(L+1) + |z|] \|f\|_\sigma. \quad (\text{RF12})$$

This estimate acts on polynomials, including when z is real.

A convenient explicit evaluation bound, slightly stronger than the received one, is

$$\sqrt{K_{\sigma,L}(z, z)} \leq E_\sigma(L, r) := \frac{e^{1+\pi r/4}}{\sqrt{M_\sigma}} (L+1)(2L+1)^{r/2+1/4}, \quad |z| \leq r. \quad (\text{RF13})$$

For $L \geq 1$, apply Cauchy's coefficient estimate to the original generating function $(1+t^2)^{-1/4} e^{z \arctan t}$ on $|t| = L/(L+1)$. The factors are bounded by $(L+1)^{1/2}(2L+1)^{-1/4}$, $e^{\pi r/4}(2L+1)^{r/2}$, and e . The orthonormal coefficient ratio is $\rho_n = \sqrt{n!/(1/2)_n} \leq \sqrt{2n+1}$. Summing $L+1$ squared bounds gives RF13. The case $L=0$ follows from $K_{\sigma,0} = 1/M_\sigma$. Since $(L+1)(2L+1)^{1/4} \leq (L+1)^{3/4}(2L+1)^{1/2}$, the received constant (18) follows as well.

Let $D_z f = (f - f(z))/(y - z)$. Evaluation and RF12 give, for $L \geq 1$,

$$\|D_z f\|_\sigma \leq \mathcal{D}(L, r) \|f\|_\sigma, \quad \mathcal{D}(L, r) = 2b_L[1 + \sqrt{M_\sigma} E_\sigma(L, r)] \quad (|z| \leq r, \deg f \leq L). \quad (\text{RF14})$$

Successive divisions by $\lambda_1, \dots, \lambda_d$, retaining their order and multiplicity, have final quotient exactly $\Omega_p f$. Expanding each identity $f = (y - \lambda_1)D_{\lambda_1} f + f(\lambda_1)$ proves this by induction. Thus

$$\|\Omega_p f\|_\sigma \leq \mathcal{D}(L, R)^d \|f\|_\sigma \quad (\deg f \leq L, L \geq d). \quad (\text{RF15})$$

If $L < d$, the quotient is zero. No rational source integral occurs.

4. An explicit polynomial lift and its native upper norm

Set

$$n_0 = 2 \left\lfloor \frac{N+1-q}{2} \right\rfloor, \quad K_0 = K_{Q,n_0}(0,0), \quad T = T_{k,N} = \frac{1}{|Q(0)|\sqrt{K_0}}, \quad H(y) = K_{Q,n_0}(y,0)/K_0. \quad (\text{RF16})$$

The finite guards used in this section are

$$q \geq 16, \quad R_k \leq 2^{-13}q, \quad n_0 \leq 2q - 2, \quad a_\Gamma^2 \leq \beta_q/2, \quad \beta_t = 2^{-27}t^2. \quad (\text{RF17})$$

IVO9-11 gives $\|QH\|_\sigma = K_0^{-1/2}$, $H(0) = 1$, and $|H(z)| \geq e^{-n_0 a_\Gamma^2 / \beta_q}$ throughout $|z| \leq a_\Gamma$. In particular its reciprocal is holomorphic on the required contour region.

For $\deg a < d$, put

$$h = -H\mathcal{R}_p\left(\frac{a}{QH}\right), \quad f = \frac{a + Qh}{p}. \quad (\text{RF18})$$

The numerator has the full required zero multiplicities at every root of p . Thus f is a polynomial and $\deg f \leq q + n_0 - 1 \leq N$. Its original class is $U_p a$. Since $\deg a < d$, it also equals $\Omega_p(Qh)$.

Define

$$A_{\Gamma,d} = \left(\sum_{j=0}^{d-1} a_\Gamma^{2j} \right)^{1/2}, \quad V_{d,N} = \left(\sum_{j=0}^{d-1} [2(N+d+1)]^{2j} \right)^{1/2},$$

$$\mathcal{U}_{p,N}^{\text{unif}} = C_d A_{\Gamma,d} M_k^- e^{n_0 a_\Gamma^2 / \beta_q} V_{d,N} \mathcal{D}(N+d, R)^d. \quad (\text{RF19})$$

RF8 bounds the coefficient norm of the remainder in RF18. Multiplying QH by its degree- $< d$ remainder, RF12 bounds each power by the corresponding term in $V_{d,N}$; all intermediate degrees are at most $N + d$. Finally use RF15 on Qh . Therefore

$$\|U_p a\|_{G_N} \leq \sqrt{u_k} \mathcal{U}_{p,N}^{\text{unif}} T \|\text{coeff } a\|_2. \quad (\text{RF20})$$

The notation records a uniform constant: its displayed value depends on d and the fixed compact set, not on individual pole separations.

5. The full rational conductor and its explicit inverse

Retain every nonzero original conductor coefficient a_j , physical shift b_j , number J , and first symbol moment $\mu_v \neq 0$. Set

$$\tau_j = (b_j - 4)/i, \quad \tilde{\mathcal{T}}_A f(y) = \sum_{j=1}^J a_j f(y + \tau_j), \quad \tilde{E}_A(z) = \sum_j a_j e^{\tau_j z}. \quad (\text{RF21})$$

This is exactly IVO1: $\tau_j = -\zeta_j$. Its first nonzero derivative is $i^{-v} \mu_v$, and $0 \leq v \leq J - 1$. Every τ_j is an even lattice translation, so $\mathcal{L} + \tau_j = \mathcal{L}$. Consequently all roots of every $p(y + \tau_j)$ remain outside the full lower lattice, even when translated poles collide.

Write $Q_- = Q_{k-8}$, $q_- = (k-7)^2$, $\Delta = q - q_- = 16k - 48$, $Z = \max_j |\tau_j|$, $A_1 = \sum_j |a_j|$, and

$$B_p(y) = \prod_{j=1}^J p(y + \tau_j), \quad L = Jd, \quad R_B = R + Z, \quad N_a(y) = B_p(y) \tilde{\mathcal{T}}_A(a/p)(y). \quad (\text{RF22})$$

Here B_p denotes only the scalar denominator polynomial; the bounded inverse operator will be denoted $\mathbf{B}_p$.

If $a/p = c y^{-n} + O(y^{-n-1})$, $n = d - \deg a \geq 1$, then

$$\tilde{\mathcal{T}}_A(a/p) = (-1)^v \binom{n+v-1}{v} i^{-v} \mu_v c y^{-n-v} + O(y^{-n-v-1}). \quad (\text{RF23})$$

Expanding $(y + \tau)^{-n}$ proves this coefficient; lower moments vanish, including every possible contribution from later Laurent coefficients. Therefore $\deg N_a = L - n - v \leq L - v - 1$, and the conductor on proper rational functions is injective.

Here is a complete finite inverse bound, without selecting a residue. Put

$$e(z) = z^{-v} \tilde{E}_A(-z), \quad e(0) = (-1)^v i^{-v} \mu_v / v!, \quad H_* = \frac{2v!}{|\mu_v|}.$$

A concrete admissible positive radius is

$$\varrho = \min \left(1, \frac{|e(0)|(v+1)!}{2A_1 \max(1, Z)^{v+1} e^Z} \right). \quad (\text{RF24})$$

The exponential-series tail satisfies $|e(z) - e(0)| \leq A_1 Z^{v+1} |z| e^{|z|} / (v+1)! \leq |e(0)|/2$ on this disk. Hence Cauchy's formula gives $||z^j]e^{-1}| \leq H_* \varrho^{-j}$.

Define

$$B_{d,L} = \sqrt{L} \sum_{j=0}^{v+d-1} \binom{L+j-1}{j} R_B^j, \quad A_d = \max_{1 \leq n \leq d} (n-1)! \sum_{\ell=1}^n \frac{H_* \varrho^{-(n-\ell)}}{(v+\ell-1)!},$$

$$\mathcal{W}_d = \sqrt{d}(1+R)^d A_d B_{d,L}. \quad (\text{RF25})$$

To prove the bound, write $a/p = \sum_{n \geq 1} c_n y^{-n}$, $\tilde{T}_A(a/p) = \sum_{n \geq 1} b_n y^{-n}$. Their exponential coefficient series obey exactly

$$\sum_{n \geq 1} \frac{b_n z^{n-1}}{(n-1)!} = \tilde{E}_A(-z) \sum_{n \geq 1} \frac{c_n z^{n-1}}{(n-1)!}. \quad (\text{RF26})$$

The first $v+d$ Laurent coefficients of N_a/B_p are bounded by $B_{d,L} \|\text{coeff } N_a\|_2$: expand $\prod_{\nu=1}^L (1 - \xi_\nu/y)^{-1}$, whose degree- j coefficient is at most $\binom{L+j-1}{j} R_B^j$. Inverting RF26 bounds each c_n , $n \leq d$, by A_d times that number. Finally multiply the first d Laurent coefficients by p and take the polynomial part. Its coefficient operator norm is at most $(1+R)^d$, by the coefficient one-norm of p . Thus

$$\|\text{coeff } a\|_2 \leq \mathcal{W}_d \|\text{coeff } N_a\|_2. \quad (\text{RF27})$$

In particular $\log^+ \mathcal{W}_d = O_{A,\mathcal{P}}(d \log(d+2))$, directly from the factorial and binomial bounds in RF25.

The exact lower quotient map is

$$M_{B_p}^{(-)} C_A U_p a = [N_a]_{Q_-}. \quad (\text{RF28})$$

Each shifted p is a unit there, by shifting a Bézout identity for p, Q and using $Q_- \mid Q(y + \tau_j)$. This proves the identity on the complete lower quotient. On the finite algebraic guard

$$L - v \leq q_-, \quad (\text{RF29})$$

every N_a is already its unreduced lower remainder. RF27 proves $C_A U_p$ injective, including all translated collisions. The degree bound is attained by $a = y^{d-1}$, so RF29 is the sharp guard for this unreduced-numerator recovery.

6. Lower norm with the full numerator and boundary product

Every polynomial f_- , of degree at most $N - v$, representing $C_A U_p a$ satisfies

$$B_p f_- - N_a = Q_- h_-, \quad \deg h_- \leq n_- := N - v + L - q_-. \quad (\text{RF30})$$

Since $\deg N_a < L$, taking the entire remainder modulo B_p gives $N_a = -\mathcal{R}_{B_p}(Q_-h_-)$. Use the circle of radius $r_B = R_B + 1$, on which every denominator factor has modulus at least one. RF8, now with L roots, gives the constant

$$C_B = r_B \sqrt{L} (R_B + r_B)^L.$$

IVO10 and RF10 give

$$\begin{aligned} \|\text{coeff } N_a\|_2 &\leq C_B |Q_-(0)| M_{k-8}^+(r_B) E_-(r_B) \sqrt{K_{Q_-,n_-}(0,0)} \|Q_-h_-\|_\sigma, \\ E_-(r_B) &= e^{n_- r_B^2 / (2\beta_{q_-})} \sqrt{1 + r_B^2 / \beta_{q_-}}. \end{aligned} \quad (\text{RF31})$$

The circle may enclose lattice roots: Q_-h_- is a polynomial, so no reciprocal of Q_- is used here.

The exact central-grid factorization is $Q = Q_- D_\partial$. Set

$$\begin{aligned} W_N &= 2(q + n_0 + 1) + R_k, \quad U_- = 2(q_- + n_- + 2), \quad V_- = b_{q_-+n_-+1} + q_-/\delta, \\ D_- &= \prod_{t=1}^{\lfloor n_-/2 \rfloor - n_0/2} \left[1 + \left(\frac{U_- V_-}{n_0 + 2t} \right)^2 \right], \quad \mathcal{R}_N = \frac{|D_\partial(0)|}{W_N^\Delta \sqrt{D_-}}. \end{aligned} \quad (\text{RF32})$$

The product is empty and equals one if its upper limit is zero. Multiplication by every boundary factor and IVO12 prove

$$\frac{1}{|Q_-(0)| \sqrt{K_{Q_-,n_-}(0,0)}} \geq \mathcal{R}_N T. \quad (\text{RF33})$$

Indeed $K_{Q_-,n_0} \leq W_N^{2\Delta} K_{Q,n_0}$, then $K_{Q_-,n_-} \leq D_- K_{Q_-,n_0}$. All factors are retained. Moreover $|D_\partial(0)| \geq [\delta(k-6)]^\Delta$, since every boundary root has an odd coordinate of modulus at least $k-6$.

Define

$$\begin{aligned} \mathcal{L}_N &= \frac{\mathcal{R}_N}{W_d C_B M_{k-8}^+(r_B) E_-(r_B)}, \quad \mathcal{B}_N = 2(N - v + L + 1) + R_B, \\ \mathcal{A}_d &= \sqrt{M_\sigma} A_1 \sqrt{d} [2(L + 1) + R_B]^{L-1}. \end{aligned} \quad (\text{RF34})$$

RF27 and RF31-33 give $\|Q_-h_-\|_\sigma \geq \mathcal{L}_N T \|\text{coeff } a\|_2$. The entire numerator in RF22 satisfies

$$\|N_a\|_\sigma \leq \mathcal{A}_d \|\text{coeff } a\|_2, \quad \|B_p f_-\|_\sigma \leq \mathcal{B}_N^L \|f_-\|_\sigma. \quad (\text{RF35})$$

For the first bound, expand each $a(y + \tau_j)$ in its original monomials. Every term is a product of at most $L-1$ factors with root modulus at most R_B ; apply RF12 to each and then $\sum_{\nu < d} |a_\nu| \leq \sqrt{d} \|a\|_2$. The second bound applies the same estimate to all L denominator factors.

For a numerical finite lower bound on T , retain IVO23:

$$T^{\text{lb}} = \frac{\sqrt{c_\sigma q / (1 + 16q^2)} (q - R_k)^q e^{-\pi q/2}}{|Q(0)| (n_0 + 1) 10^{n_0}}, \quad c_\sigma = \Gamma(1/4)^2 / (2\pi). \quad (\text{RF36})$$

Its proof is the full Legendre evaluation bound on $[q, 2q]$, the Gamma lower envelope, and every factor of Q . Thus $T \geq T^{\text{lb}}$.

The complete finite domain for the rational norm theorem is

$$\boxed{q_- \geq 16, \quad R_k \leq 2^{-14} q_-, \quad a_\Gamma^2 \leq \beta_q/2, \quad L - v \leq q_-, \quad n_- + 2 \leq 2q_-, \quad \mathcal{L}_N T^{\text{lb}} \geq 2\mathcal{A}_d.} \quad (\text{RF37})$$

These imply RF17 and every required relation-kernel degree: $N - q - n_0$ is either -1 or 0 , and $L - v = Jd - v \geq 1$, so

$$n_- - n_0 = \Delta + L - v + (N - q - n_0) \geq \Delta > 0.$$

Thus $0 \leq n_0 \leq n_- \leq 2q_- - 2 \leq 2q - 2$. The radii of both Q and Q_- are at most $R_k \leq 2^{-14}q_-$, within the respective 2^{-13} root-radius allowances in IVO9. IVO10 permits every complex evaluation radius and retains its displayed exponential; no extra inequality involving r_B is needed in RF31. The separate bound $a_{\Gamma}^2 \leq \beta_q/2$ is required and imposed for the nonvanishing reciprocal H^{-1} in IVO11. The last inequality in RF37 absorbs the complete numerator in RF30, giving for every representative

$$\|f_-\|_{\sigma} \geq \frac{\mathcal{L}_N}{2\mathcal{B}_N^L} T \|a\|_2. \quad (\text{RF38})$$

To pass through the entire observation, use the exact original inclusion $K \subseteq \ker C_A$. Therefore there is a well-defined map $\widehat{C}_A : B \rightarrow E_-$ with $\widehat{C}_A \Lambda = C_A$. Every lift in RF4 has the same conductor image. IVO4, proved by applying the complete conductor to each such lift, gives

$$\|P_B U_p a\|_{G_N} \geq \frac{\sqrt{\ell_{k,N}}}{\mathcal{F}_N} \|C_A U_p a\|_{\sigma, Q_-, N-v}, \quad \mathcal{F}_N = A_1 e^{1+2\pi\gamma} (N+1)(2N+1)^{4\delta+1/2}. \quad (\text{RF39})$$

Taking the minimum in RF38 now proves

$$\boxed{\ell_{k,N} \left(\frac{\mathcal{L}_N}{2\mathcal{F}_N \mathcal{B}_N^L} \right)^2 T^2 I_d \preceq H_{p,N}^B \preceq H_{p,N} \preceq u_k (\mathcal{U}_{p,N}^{\text{unif}})^2 T^2 I_d.} \quad (\text{RF40})$$

This proof does not identify the conductor kernel with K .

Let $E_{k,d,N}$ be the maximum of zero, the negative logarithm of the left scalar in RF40 after division by $u_k T^2$, and the logarithm of the right scalar after that division. All its constants are displayed. For fixed original data,

$$E_{k,d,N} = O_{h,A,\mathcal{P}}((k + Jd) \log(q + Jd + 2)). \quad (\text{RF41})$$

The bounds follow from RF8-15, RF25 and the $O(k + Jd)$ factors in RF32. No contour constant depends on pole collisions. If $d \log(q + 2) = o(q)$, then $d = o(q)$, $n_- = N - q + O(k + Jd)$, and all degree/radius guards in RF37 hold eventually, uniformly over the pole family and $q-1 \leq N \leq 2q$. Also $\log T^{\text{lb}} \geq q \log(q/k) - O_h(q)$, whereas $\log^+(\mathcal{A}_d/\mathcal{L}_N) = o(q)$, so the absorption guard holds eventually. Consequently RF40 proves received (12)-(13) with $E_{k,d} = \max_N E_{k,d,N} = o(q)$.

7. The remainder projection is onto and the inverse has rank d at the late scale

Define

$$T_p = \mathcal{R}_p L_N, \quad \mathbf{B}_p = J_N \mathfrak{Q}_p L_N, \quad F_p = U_p T_p.$$

Complete polynomial division of the actual minimum representative gives

$$X = p(M)^{-1} = \mathbf{B}_p + F_p, \quad \|\mathbf{B}_p\|_{G_N} \leq \beta_N := \sqrt{u_k/\ell_{k,N}} \mathcal{D}(N, R)^d. \quad (\text{RF42})$$

When $N < d$, the quotient vanishes and β_N may be set to zero. In the eventual growing family $d < N$. In all cases $\log^+ \beta_N = O_{h,\mathcal{P}}(k + d \log(q + 2))$.

For the low-polynomial inclusion $\iota : a \mapsto [a]$, write its original minimum lift as $L_N[a] = a - Qh_a$. The second term is the μ_k -orthogonal projection of a onto the entire relation space $Q\mathcal{P}_{N-q}$. Hence

$$\|Qh_a\|_\sigma \leq \sqrt{u_k/\ell_{k,N}} \|a\|_\sigma. \quad (\text{RF43})$$

If $N = q - 1$, this term is zero. Otherwise $\deg h_a \leq N - q \leq n_0$. Define

$$P_d = \sqrt{M_\sigma} \left(\sum_{j=0}^{d-1} [2(d+1)]^{2j} \right)^{1/2}, \quad E_0(r) = e^{n_0 r^2 / (2\beta_q)} \sqrt{1 + r^2 / \beta_q},$$

$$\eta_N = \frac{C_d M_k^+(a_\Gamma) E_0(a_\Gamma) \sqrt{u_k/\ell_{k,N}} P_d}{T^{\text{lb}}}. \quad (\text{RF44})$$

At $N = q - 1$ set $\eta_N = 0$. The coefficient bound $\|a\|_\sigma \leq P_d \|a\|_2$, RF8, RF10, and IVO10 show

$$T_p \iota = I - \mathcal{E}_{p,N}, \quad \|\mathcal{E}_{p,N}\| \leq \eta_N. \quad (\text{RF45})$$

Here the factor $|Q(0)|\sqrt{K_0} = T^{-1}$ arises exactly; no small relation projection is presumed. Thus the finite guard $\eta_N < 1$ proves that $\iota(I - \mathcal{E}_{p,N})^{-1}$ is a right inverse of T_p , of norm at most

$$\sqrt{u_k} P_d / (1 - \eta_N). \quad (\text{RF46})$$

Also RF8 and RF13 on the actual minimum section give

$$\|T_p\| \leq C_d E_\sigma(N, a_\Gamma) / \sqrt{\ell_{k,N}}. \quad (\text{RF47})$$

For an onto Hilbert-space map, the norm of any right inverse bounds the reciprocal of its least positive singular value: apply its adjoint to a target unit vector, or use the attained inverse. Combining RF40 and RF46–47 gives

$$f_- T \leq s_d(F_p) \leq s_1(F_p) \leq f_+ T, \quad (\text{RF48})$$

$$f_- = \sqrt{\ell_{k,N}/u_k} \frac{\mathcal{L}_N(1 - \eta_N)}{2\mathcal{F}_N \mathcal{B}_N^L P_d}, \quad f_+ = \sqrt{u_k/\ell_{k,N}} \mathcal{U}_{p,N}^{\text{unif}} C_d E_\sigma(N, a_\Gamma).$$

This proves rank d ; rank is not inferred from the division identity alone.

A complete useful cluster guard is

$$\boxed{\eta_N \leq \frac{1}{2}, \quad f_- T^{\text{lb}} > 2\beta_N.} \quad (\text{RF49})$$

Both follow eventually from the explicit growth in RF36, uniformly over the admissible pole family. Weyl's singular-value inequality and the rank- d perturbation bound give

$$f_- T - \beta_N \leq s_d(X) \leq s_1(X) \leq f_+ T + \beta_N, \quad s_{d+1}(X) \leq \beta_N \quad (d < q). \quad (\text{RF50})$$

For completeness the last inequality follows by restricting X to $\ker F_p$, of dimension at least $q - d$, where $X = \mathbf{B}_p$. Taking reciprocal singular values in the original metric therefore proves

$$\log s_{q-d+j,N}(p(M)) = -\log T + O_{h,A,\mathcal{P}}((k + Jd) \log(q + Jd + 2)), \quad 1 \leq j \leq d,$$

$$s_{q-d,N}(p(M)) \geq \beta_N^{-1} \quad (d < q). \quad (\text{RF51})$$

The least d singular directions are exactly the late cluster because their logarithms are $-q \log(q/k) + O_h(q)$, while the remaining ones are bounded below by $e^{-o(q)}$.

8. Evaluated exponent and the complete observed late heat

The actual retained analytic providers are H4-9 of the original profile proof, S2-4 and S22-30 of the smallest-singular proof, and IFH8-10 of the original flag proof. Their profile and zero-law status is retained, rather than reclassified as a finite matrix computation.

Here is the exact connection to T . For even $n = n_0$, let V_n, ν_n^Γ be the monic polynomial and its squared norm for the full $Q^2 d\sigma$. Parity and Christoffel-Darboux give

$$K_{Q,n}(0,0) = \frac{V_n(0)^2 \Xi_n}{\nu_n^\Gamma}, \quad \Xi_n = \frac{V'_{n+1}(0)}{V_n(0)}. \quad (\text{RF52})$$

Positive-zero interlacing as in IFH9 gives $1 \leq \Xi_n \leq 4(q+n+2)^2/\beta_q$; $\Xi_0 = 1$. Thus $\log \Xi_n = O(1)$ on the retained degree window. Applying NG2 on the full degree- $q+n$ monic source and relation fibres transfers H4 to the Gamma ratio: each of the two minima changes by a factor between the same positive source-comparison constants, so the logarithm of their ratio changes by at most $\log(u_k/\ell_{k,2q+2}) = O_h(k+\log q)$. Therefore the full reference monic profile gives $\log(\nu_n^\Gamma/\gamma_{q+n}) = 2q\psi(n/q) + o(q)$, where $\gamma_j = M_\sigma j!(1/2)_j$. The zero law, with the finite IVO9 gap permitting the logarithmic test, gives

$$\log |V_n(0)| = n \log q + \frac{sq}{2} \int \log x \rho_s(x) dx + o(q), \quad n/q \rightarrow s > 0.$$

At $s = 0$, the fixed scaled lower and upper zero bounds make the residual $O(n) = o(q)$. The actual root product, proved with its $O(k)$ error in HAR35 of the pinned harmonic proof, is

$$\log |Q(0)| = q \log k + \frac{q}{2} I + O_{\delta,\gamma}(k), \quad I = \log(\delta^2 + \gamma^2) - 3 + \frac{\delta}{\gamma} \arctan \frac{\gamma}{\delta} + \frac{\gamma}{\delta} \arctan \frac{\delta}{\gamma}.$$

Substituting into RF52 proves on the established window $0 \leq s \leq 1$, including each adjacent-cutoff pair,

$$\log T = q \log(q/k) + q[\psi(s) + J(s) - I/2] + o_h(q),$$

$$J(s) = (1+s)(\log(1+s) - 1) - \frac{s}{2} \int \log x \rho_s(x) dx, \quad J(0) = -1. \quad (\text{RF53})$$

No profile outside this source window is needed or claimed. RF51 and RF53 prove received (26)–(27), uniformly in the admissible pole family and singular index.

At the four actual cutoffs $q-1, q, 2q-1, 2q$, use the same polynomial and signs $+, +, -, -$. Put

$$\mathfrak{C} = 4[\psi(0) - \psi(1) - J(1) - 1], \quad 1.353893 < \mathfrak{C} < 1.354751.$$

The interval is the inherited directed certificate, not a new numerical test. Taking determinants of RF40, and observing that u_k and the numerator coefficient frame are identical at all endpoints, gives

$$\begin{aligned} \mathcal{R} \log \det H_{p,N}^X &= \mathfrak{C} dq + o(dq), \quad X = E, B, \\ \mathcal{R} \sum_{j=q-d+1}^q \log s_{j,N}(p(M))^2 &= -\mathfrak{C} dq + o(dq). \end{aligned} \quad (\text{RF54})$$

The finite determinant error is bounded by $d \sum_N E_{k,d,N}$.

For the common time

$$\tau_k^* = \exp\{2q[\log(q/k) - 1 - I/2]\}, \quad (\text{RF55})$$

every slow heat exponent tends to zero exponentially at the low pair, because $-1 - \psi(0) - J(0) = -\psi(0) < 0$. At the high pair it tends to infinity exponentially, because $\psi(1) < 1/8$ and the retained $J(1) < -1.163427$ imply $-1 - \psi(1) - J(1) > 0$. The remaining heat contribution is at most $(q - d)e^{-\tau_k^*/\beta_N^2}$, by RF50. Hence

$$\frac{1}{d} \mathcal{R} \text{Tr} e^{-\tau_k^* p(M)^\dagger p(M)} \longrightarrow 2. \quad (\text{RF56})$$

The observed assertion needs the actual original projection, rather than unweighted slow counts. Let V_d be the top left singular space of X . If $x \in V_d$, then $\|X^{-1}x\| \leq \|x\|/s_d(X)$, and

$$\|(I - P_{\text{ran } U_p})x\| = \|(I - P_{\text{ran } U_p})\mathbf{B}_p X^{-1}x\| \leq \epsilon_N \|x\|, \quad \epsilon_N = \frac{\beta_N}{f_- T - \beta_N}. \quad (\text{RF57})$$

RF40 gives $\ell_N = e^{-2E_{k,d,N}}$ on $\text{ran } U_p$. For any orthogonal projections onto K , U , and any unit x at distance at most ϵ from U , Cauchy-Schwarz in the two orthogonal components gives

$$|\langle z, x \rangle| \leq a\sqrt{1-t^2} + \sqrt{1-a^2}t, \quad a = \|P_U z\| \leq \sqrt{1-\ell}, \quad t = \|(I - P_U)x\| \leq \epsilon, \quad z \in K, \quad \|z\| = 1.$$

If $\epsilon < \sqrt{\ell}$, this expression increases in both a, t on these intervals. Substituting their endpoints and subtracting the square from one proves the sharp lower fraction

$$\ell_{\text{sharp},N} = \left(\sqrt{\ell_N} \sqrt{1 - \epsilon_N^2} - \sqrt{1 - \ell_N \epsilon_N} \right)_+^2. \quad (\text{RF58})$$

For $\epsilon_N \geq \sqrt{\ell_N}$ this bound is zero. If needed use $\min(1, \epsilon_N)$ in the displayed expression. The finite additional guard $\epsilon_N \leq \sqrt{\ell_N}/4$ implies $\ell_{\text{sharp},N} \geq \ell_N/4$. It holds eventually because $\log \epsilon_N = -q \log(q/k) + O_h(q)$, whereas $E_{k,d,N} = o(q)$.

If $Xu'_j = s_j(X)u_j$, then $p(M)^\dagger p(M)u_j = s_j(X)^{-2}u_j$. Thus the same left singular vectors are exactly the slow heat eigenvectors. Their observed weights are $\|P_B u_j\|^2$, bounded below by RF58. The measured trace is

$$\text{Tr}(\Lambda e^{-\tau p(M)^\dagger p(M)} L_{B,N}) = \sum_j e^{-\tau/s_j(X)^2} \|P_B u_j\|^2. \quad (\text{RF59})$$

At $\tau = \tau_k^*$, the positive low pair contributes at least $d(\ell_{\text{sharp},q-1} + \ell_{\text{sharp},q})(1 - o(1))$. The total negative high pair is at most $2q \exp[-\exp(cq)]$ for some fixed $c > 0$, eventually; all noncluster tails satisfy the same type of stronger bound. The first term is $de^{-o(q)}$, so it dominates. The upper bound on the return divided by d is $2 + o(1)$. Consequently the return is eventually positive and

$$\frac{1}{q} \log \left[\frac{1}{d} \mathcal{R} \text{Tr}(\Lambda e^{-\tau_k^* p(M)^\dagger p(M)} L_{B,N}) \right] \longrightarrow 0. \quad (\text{RF60})$$

For a nonmonic filter $c_k p_k$, singular values are multiplied by $|c_k|$ and heat time by $|c_k|^2$, exactly.

9. The explicit ES datum, its action, and the original kernel quotient

The selected witness and filter datum are attributed to the received continuation; no unread ES foundation file is claimed as a source. Direct rational arithmetic proves

$$\begin{aligned} \frac{4}{13} &= \frac{1}{4} + \frac{1}{18} + \frac{1}{468}, \\ P_{\text{ES}}(t) &= (t-13)(t-4)(t-18)(t-468) \\ &= t^4 - 503t^3 + 16738t^2 - 168480t + 438048, \\ \sum_{\lambda=13,4,18,468} \lambda^{-1} &= \frac{5}{13}. \end{aligned} \tag{RF61}$$

Take the literal polynomial $p_k = P_{\text{ES}}^k$, degree $d = 4k$. Its real poles have distance at least δ from every original odd-lattice root. Since $4k \log(q+2) = o(q)$, all preceding eventual conclusions apply with every repetition retained:

$$\begin{aligned} \mathcal{R} \log \det H_{p_k, N}^B &= 4\mathfrak{C} kq + o(kq), \quad 5.415572 < 4\mathfrak{C} < 5.419004, \\ \frac{1}{4k} \mathcal{R} \text{Tr} e^{-\tau_k^* p_k(M)^\dagger p_k(M)} &\longrightarrow 2, \end{aligned} \tag{RF62}$$

and the original measured heat obeys RF60. For $J \leq 81$, the unreduced-conductor guard follows from $324k \leq (k-7)^2$ when $k \geq 341$, $k \equiv 1 \pmod{4}$: the difference $k^2 - 338k + 49$ is 1072 at 341 and increasing thereafter. This is an algebraic threshold only. RF37 and RF49, not this single integer, specify the additional finite analytic guards.

For any admissible monic p , let C_p be multiplication by y on $\mathbb{C}[y]/(p)$ in its unchanged coefficient frame. Polynomial division of ya gives the exact maps

$$MU_p = U_p C_p + [1]e_{d-1}^*, \quad M_S U_p = U_p(kI/2 + iC_p) + i[1]e_{d-1}^*. \tag{RF63}$$

The quotient ya/p has constant part equal to the leading coefficient of a , proving both identities. The attained compression is

$$H_{p, N}^{-1} U_p^* G_N M U_p = C_p + H_{p, N}^{-1} U_p^* G_N [1]e_{d-1}^*. \tag{RF64}$$

The domain of U_p has its pulled-back metric $H_{p, N}$. If its domain instead uses the Euclidean metric, its adjoint is $U_p^* G_N$, not $H_{p, N}^{-1} U_p^* G_N$. This explicit metric convention repairs the ambiguity in the received adjoint notation (57).

The exact monic physical denominator is $D(S) = i^d p((S - k/2)/i)$, with numerator map $a \mapsto i^d a((S - k/2)/i)$. The quotient-coordinate isomorphism $[f(y)] \mapsto [f((S - k/2)/i)]$ intertwines these two rational maps, since both i^d factors cancel in their ratio. The source and observed Grams transform by the same numerator congruence. If an existing physical unit is subsequently multiplied, it acts on every term of RF63, including the defect vector. The ES integers are filter poles; these exact maps do not identify them with zeta zeros.

Finally define, in the original coefficient coordinates,

$$\pi_p[x] = \mathfrak{Q}_p(\text{rem}_Q(px)) \in \mathcal{P}_{< q-d}. \tag{RF65}$$

For x of degree below $q - d$, px has degree below q , so $\pi_p[x] = x$. Thus the low-polynomial inclusion is a right inverse. Also $\pi_p[x] = 0$ exactly when $\text{rem}_Q(px) = a$ has degree below d , equivalently $[x] = U_p a$. Hence $\ker \pi_p = \text{ran } U_p$. RF40 implies this subspace meets K only at zero, so $\pi_p|_K$ is injective.

Let I_K be any fixed full frame. Completing the quadratic form over the entire U_p -fibre proves that the attained quotient metric is $(\pi_p G_N^{-1} \pi_p^*)^{-1}$. The two Schur complements of the full Gram of $[I_K, U_p]$ then give the exact equality

$$\begin{aligned} \delta_{p,N} &= \log \det H_{p,N} - \log \det H_{p,N}^B \\ &= \log \det(I_K^* G_N I_K) - \log \det[(\pi_p I_K)^* (\pi_p G_N^{-1} \pi_p^*)^{-1} (\pi_p I_K)] . \end{aligned} \quad (\text{RF66})$$

Indeed, after using orthonormal frames of K, U_p , both ratios are determinants of $I - C^* C$ or $I - C C^*$, where C is their full cross Gram. Their nonzero singular values coincide and their number is at most $\min(\dim K, d)$. RF40 bounds each observed relative eigenvalue below by $e^{-2E_{k,d,N}}$. Therefore

$$0 \leq \delta_{p,N} \leq 2 \min(\dim K, d) E_{k,d,N}. \quad (\text{RF67})$$

On the actual original kernel, $\dim K = 8k - 16$, so RF41 gives $\sum_N \delta_{p,N} = o(kq)$. Subtracting RF66 at the same four cutoffs proves received (58) with the complete original kernel and attained quotient intact. The result preserves its leading coefficient; it does not assign a value to that coefficient.

10. What the finite theorem establishes

RF37 is the complete finite norm domain. RF44 and RF49 make the remainder map onto and separate exactly d singular directions. RF58 states the further explicit observation guard for transferring the cluster. All of these follow from $d \log(q+2) = o(q)$ with fixed original data, but their finite inequalities remain available before passage to a limit.

The original-source scalar factors, the full conductor numerator, every boundary root, the physical i in the coordinate map, the pulled-back adjoint metric, and the whole observation minimum have appeared explicitly. The retained high-endpoint coefficient uses the established H/IFH analytic inputs. The separate finite reduced model and complete memory/energy calculations are handled by their own derivations, not replaced by this rational norm theorem.

The independent checker `check_rational_filters.py` records 113 passing exact checks in `RATIONAL_FILTER_CHECKS.json`. It computes Hermite contour residues at repeated poles, the complete Laurent inverse with translated-pole collisions, both actual lattice shifts, a full polynomial moment source with its entire relation minimum, a complex observation and all its cross terms, the action defect, both quotient angle determinant ratios, and the ES arithmetic. Its small lattices test exact algebra; they are explicitly outside the asserted eventual analytic domain. No new native moment computation or limiting-profile certificate is claimed.

Part IV

Observed reduced model, repeated poles and slow energy

The full observed polynomial action: reduced resolvents, heat, and exact pole masses

This derivation proves the finite statements (33)–(46) in the received memory/filter continuation and extends the pole and energy-graph statements to every eigenvalue below the kernel block, with multiplicities retained. All adjoints, norms, projections and minima use the original metric. The source-family and canonical memory estimates are proved in the accompanying derivations.

1. Original spaces and the exact finite model

Fix an original cutoff. Let (E, G) have dimension q , let $\Lambda : E \rightarrow B$ be the original onto observation, and put

$$Q_B = (\Lambda G^{-1} \Lambda^*)^{-1}, \quad L_B = G^{-1} \Lambda^* Q_B. \quad (\text{RM1})$$

Then $\Lambda L_B = I_B$, $L_B^\dagger = \Lambda$, and L_B is an isometry. Indeed $L_B^* G L_B = Q_B$; completing the quadratic minimum proves that this is the entire observed quotient metric. The projection onto $K^\perp$, where $K = \ker \Lambda$, is $P_B = L_B \Lambda$.

For an admitted monic filter p of degree d , retain

$$X = p(M)^{-1} = B_p + U_p T_p, \quad U_p a = [a/p], \quad T_p = \text{rem}_p L_N, \quad (\text{RM2})$$

where L_N is the original source-minimum section. The polynomial quotient identity proves RM2: if $L_N x = p \mathfrak{Q}_p L_N x + \text{rem}_p L_N x$, then applying the original remainder map gives $x = p(M) B_p x + T_p x$, and division by $p(M)$ gives the result. Thus no independent compressed action is substituted.

Set

$$H_p = U_p^* G U_p, \quad V = U_p H_p^{-1/2}, \quad F = U_p T_p, \quad K = H_p^{1/2} T_p G^{-1} T_p^* H_p^{1/2}, \quad (\text{RM3})$$

$$Y = \Lambda V, \quad \Theta = Y^\dagger Y = H_p^{-1/2} U_p^* \Lambda^* Q_B \Lambda U_p H_p^{-1/2}. \quad (\text{RM4})$$

These are exact operators on the coefficient space, equipped here with its stated Euclidean metric. The source and target changes in RM3–4 are isometries with their original metrics; no physical coordinate or source mass is changed. Direct multiplication proves

$$V^\dagger V = I_d, \quad F F^\dagger = V K V^\dagger, \quad 0 \preceq \Theta \preceq I_d. \quad (\text{RM5})$$

The admitted rational-source estimate proves that T_p is onto and ΛU_p is injective on its displayed finite guards. Thus $K > 0$ and $\Theta > 0$ there. Outside those guards all formulas containing their inverses have that explicit restriction; the covariance identities themselves do not require it.

Let

$$\beta \geq \|B_p\|, \quad \kappa_- = \lambda_{\min} K, \quad \kappa_+ = \|K\|, \quad \varepsilon_R = 2\beta\sqrt{\kappa_+} + \beta^2. \quad (\text{RM6})$$

Since $\|F\| = \sqrt{\kappa_+}$, expansion of $X X^\dagger$ proves

$$\|X X^\dagger - F F^\dagger\| \leq \varepsilon_R, \quad \|X X^\dagger - F F^\dagger\|_{\text{HS}} \leq 2\beta\sqrt{\text{Tr } K} + \sqrt{q}\beta^2. \quad (\text{RM7})$$

For the second bound use $\|B_p F^\dagger\|_{\text{HS}} \leq \beta \|F\|_{\text{HS}}$, its adjoint, and $\|B_p B_p^\dagger\|_{\text{HS}} \leq \sqrt{q} \beta^2$. These steps retain both cross terms. The additional guard $\kappa_- > 4\beta^2$ gives a separated cluster by singular-value min-max: the first d singular values of X are at least $\sqrt{\kappa_-} - \beta > \beta$, and every remaining one is at most β .

2. Complete resolvent and heat bounds

Put $H = p(M)^\dagger p(M) > 0$, so $H^{-1} = XX^\dagger$. For positive semidefinite A, B and $z \geq 0$, multiplication of both sides gives the exact identity

$$A(I + zA)^{-1} - B(I + zB)^{-1} = (I + zA)^{-1}(A - B)(I + zB)^{-1}. \quad (\text{RM8})$$

The two outside factors are contractions, even if A and B do not commute. Use $A = XX^\dagger$, $B = FF^\dagger$, and RM5. The function $x/(1 + zx)$ vanishes at zero, so functional calculus on $\text{im } V$ and its orthogonal complement gives

$$\left\| \Lambda(zI + H)^{-1} L_B - Y(\mathbf{K}^{-1} + zI_d)^{-1} Y^\dagger \right\| \leq \varepsilon_R \quad (z \geq 0). \quad (\text{RM9})$$

The estimate at $z = 0$ is included. It is an absolute estimate with the displayed size, not an assertion that this size tends to zero without its late-scale comparison.

For $\tau > 0$ define $f_\tau(0) = 0$, $f_\tau(x) = e^{-\tau/x}$ for $x > 0$. Its derivative is $\tau x^{-2} e^{-\tau/x}$. With $u = \tau/x$, maximizing $u^2 e^{-u}/\tau$ at $u = 2$ proves

$$\sup_{x \geq 0} |f'_\tau(x)| = \frac{4}{e^2 \tau}. \quad (\text{RM10})$$

Take orthonormal eigenbases u_i, v_j of A, B . The (i, j) matrix entries in these bases satisfy

$$\langle u_i, (f_\tau(A) - f_\tau(B)) v_j \rangle = [f_\tau(a_i) - f_\tau(b_j)] \langle u_i, v_j \rangle,$$

while the corresponding entry of $A - B$ is $(a_i - b_j) \langle u_i, v_j \rangle$. Squaring, summing and applying RM10 proves the Hilbert-Schmidt estimate. Trace norm on E is at most $\sqrt{q}$ times that norm. Compression by $\Lambda = L_B^\dagger$ and L_B does not increase trace norm. Thus

$$\|\Lambda e^{-\tau H} L_B - Y e^{-\tau \mathbf{K}^{-1}} Y^\dagger\|_1 \leq \varepsilon_H(\tau) := \frac{4}{e^2 \tau} \left[2\sqrt{q} \beta \sqrt{\text{Tr } \mathbf{K}} + q\beta^2 \right]. \quad (\text{RM11})$$

Taking traces proves, in the exact given order,

$$\left| \text{Tr}(\Lambda e^{-\tau H} L_B) - \text{Tr}(\Theta e^{-\tau \mathbf{K}^{-1}}) \right| \leq \varepsilon_H(\tau). \quad (\text{RM12})$$

There is no commutation premise on $\Theta, \mathbf{K}$. This proves the received model through all modes, including the entire complement of its d -dimensional range.

3. The actual reduced memory pencil and its mass matrix

Let $a = \lambda_{\min} \Theta > 0$, and $V_B = Y \Theta^{-1/2}$, an original-target isometry. Write

$$R_{\text{red}}(z) = V_B^\dagger \Lambda(zI + H)^{-1} L_B V_B, \quad R_0(z) = \Theta^{1/2} (\mathbf{K}^{-1} + zI)^{-1} \Theta^{1/2}. \quad (\text{RM13})$$

The two inversions in this construction first eliminate the original kernel and then the remaining observed complement; associativity of the full quadratic minimum gives precisely this restriction of the full inverse. RM9 implies $\|R_{\text{red}} - R_0\| \leq \varepsilon_R$, while

$$R_0(z) \succeq \frac{a}{\kappa_-^{-1} + z} I_d.$$

Consequently, when the explicit finite quantity

$$\delta_R(z) = \frac{\varepsilon_R(\kappa_-^{-1} + z)}{a} < 1, \quad (\text{RM14})$$

the comparison and its inverse are

$$(1 - \delta_R)R_0 \preceq R_{\text{red}} \preceq (1 + \delta_R)R_0, \\ \frac{\widehat{\mathcal{F}}_{\text{red}}(z)}{1 + \delta_R} \preceq \mathcal{F}_{\text{red}}(z) \preceq \frac{\widehat{\mathcal{F}}_{\text{red}}(z)}{1 - \delta_R}, \quad \widehat{\mathcal{F}}_{\text{red}}(z) = \Theta^{-1/2}(\mathbb{K}^{-1} + zI)\Theta^{-1/2}. \quad (\text{RM15})$$

This proves the exact relationship between the static reduced matrix and the spectral pencil. Its coefficient of z is Θ^{-1} ; the original observed mass is retained.

An immediate determinant estimate useful for the programme is

$$-d \log(1 + \delta_R) \leq \log \frac{\det \mathcal{F}_{\text{red}}}{\det \widehat{\mathcal{F}}_{\text{red}}} \leq -d \log(1 - \delta_R). \quad (\text{RM16})$$

It follows by conjugating RM15 by the positive model square root and multiplying eigenvalues. At all four cutoffs, the absolute signed error is at most the sum of these four finite bounds. The constant d is retained; it is not lost on passage from operators to determinants.

In the proved rational-source family, put $T = T_{k,N}$. Its explicit bounds give $\log T = q \log(q/k) + O_h(q)$, $\log \beta = o(q)$, $\log \kappa_{\pm} = 2 \log T + o(q)$, and $-\log a = o(q)$. Substitution in RM11 and RM14 shows

$$\varepsilon_H(\tau) = \exp[-q \log(q/k) + O_h(q)] \quad \text{if } \log \tau = 2q \log(q/k) + O_h(q), \\ \delta_R(x/T^2) = \exp[-q \log(q/k) + O_{h,A,\mathcal{P}}(q)] \quad (0 \leq x \leq x_0 < \infty). \quad (\text{RM17})$$

These displays are upper bounds; a vanishing error is allowed. Their meaning is $\varepsilon_H, \delta_R \leq \exp[-q \log(q/k) + Cq]$ eventually, with the prescribed fixed data. The explicit finite formulas RM11 and RM14 control the threshold. Since $\log d = O(\log q)$, RM16 remains exponentially small on this scale. Every error is smaller than the retained observation floor $e^{-o(q)}$. No effective threshold for a separate reference zero law is introduced.

4. Exact pole reconstruction in the original metric

The general Feshbach-Schur construction and reconstruction map are established in Geneviève Dusson, Israel Michael Sigal and Benjamin Stamm, *The Feshbach-Schur map and perturbation theory* (2021), original arXiv source, version 1, Section 1, theorem thm:isospF, equations Fesh, QP, and proposition prop:U-prop-SA; the original file was read at lines 398-490. The following finite proofs establish the constants and original-metric identities used here. They do not require the paper's separate small-perturbation theorem.

Decompose $E = K \oplus K^{\perp}$, identifying $K^{\perp}$ with the original B via RM1, and retain

$$H = \begin{pmatrix} A & B_H \\ B_H^{\dagger} & D_H \end{pmatrix} > 0, \quad \mathcal{F}(z) = zI + D_H - B_H^{\dagger}(A + zI)^{-1}B_H. \quad (\text{RM18})$$

If $A + zI$ is invertible, solve the first block equation in $(H + zI)(x, b) = (0, c)$. It gives $x = -(A + zI)^{-1}B_H b$ and then $\mathcal{F}(z)b = c$. Thus $\mathcal{F}(z)^{-1} = \Lambda(H + zI)^{-1}L_B$ whenever both inverses exist. The same elimination gives a congruence between $H + zI$ and $\text{diag}(A + zI, \mathcal{F}(z))$ for real z ; it preserves the full inertia because the elimination matrix is invertible.

Let h be a simple least eigenvalue, with unit eigenvector w , and let the remaining eigenvalues be at least $g > h$. Put $\theta = \|\Lambda w\|_{Q_B}^2 > 0$. For unit $x \in K$, $|\langle w, x \rangle|^2 \leq 1 - \theta$. The spectral theorem gives

$$A \succeq [h(1 - \theta) + g\theta]I > hI. \quad (\text{RM19})$$

In particular the expressions at $z = -h$ below are defined. Put $b = \Lambda w / \sqrt{\theta}$. The block eigenvector equations and its unit norm give

$$\mathcal{F}(-h)b = 0, \quad w = \sqrt{\theta} \begin{pmatrix} -(A - hI)^{-1} B_H b \\ b \end{pmatrix}, \quad b^\dagger \mathcal{F}'(-h)b = \theta^{-1}. \quad (\text{RM20})$$

The derivative is $I + B_H^\dagger (A - hI)^{-2} B_H$, so the last identity is exactly the norm of the displayed reconstructed vector. The spectral resolution of the full resolvent proves

$$\text{Res}_{z=-h} \mathcal{F}(z)^{-1} = \theta b b^\dagger. \quad (\text{RM21})$$

For the static Schur matrix $S = \mathcal{F}(0)$, the same resolution gives $S^{-1} = (\theta/h) b b^\dagger + R$, $0 \leq R \leq g^{-1}I$. Therefore

$$\frac{h}{\theta + h/g} \leq \lambda_{\min} S \leq \frac{h}{\theta}, \quad \lambda_2(S) \geq g \quad (\dim B \geq 2). \quad (\text{RM22})$$

The last bound follows by the max-min principle on the complement of b , where the rank-one term vanishes. RM20-22 give the exact relation among the full pole, the static eigenvalue and the observation residue.

5. Multiplicities and the entire slow cluster

Let $W : \mathbb{C}^d \rightarrow E$ be an isometry onto a slow spectral subspace, with all eigenvalues in $[h_-, h_+]$, and suppose the orthogonal complement has spectrum at least $g > h_+$. Put $\Theta_W = W^\dagger P_B W$, $a_W = \lambda_{\min} \Theta_W > 0$. Since $\|P_K W\|^2 = \|I - \Theta_W\| = 1 - a_W$, every unit $x \in K$ satisfies $\|W^\dagger x\|^2 \leq 1 - a_W$. Hence

$$A \succeq [h_-(1 - a_W) + g a_W]I \succeq g a_W I. \quad (\text{RM23})$$

This is a bound for the entire actual cluster. RF57-58 supplies an explicit value in the rational family: put $\epsilon_N = \beta_N / (f_- T - \beta_N)$, $\ell_N = e^{-2E_{k,d,N}}$, and

$$a_W \geq \ell_{\text{sharp},N} = \left(\sqrt{\ell_N} \sqrt{1 - \epsilon_N^2} - \sqrt{1 - \ell_N \epsilon_N} \right)^2 \quad \text{when } \epsilon_N \leq \sqrt{\ell_N}/4.$$

RF50 gives $g \geq \beta_N^{-2}$. Thus the computable choice $\alpha = \beta_N^{-2} \ell_{\text{sharp},N}$ is valid in RM28. The additional finite comparison $(f_- T - \beta_N)^{-2} < \alpha$ puts the entire slow cluster below the kernel block. All these inequalities hold eventually on the growing-degree family by RF36-58. The bounds give $a_W \geq e^{-o(q)}$ and $g \geq e^{-o(q)}$, with the full observed Gram retained.

For an eigenvalue $h < \lambda_{\min} A$ of multiplicity s , choose an orthonormal eigenframe W_h , put $C_h = \Lambda W_h$, and $\Theta_h = C_h^\dagger C_h$. The matrix C_h is injective: a full eigenvector in K would have energy at least $\lambda_{\min} A > h$. Thus $\Theta_h > 0$. Block reconstruction, column by column, proves

$$\begin{aligned} \mathcal{F}(-h)C_h &= 0, & C_h^\dagger \mathcal{F}'(-h)C_h &= I_s, \\ \text{Res}_{z=-h} \mathcal{F}(z)^{-1} &= C_h C_h^\dagger. \end{aligned} \quad (\text{RM24})$$

In the isometric observed frame $Y_h = C_h \Theta_h^{-1/2}$, these become

$$Y_h^\dagger \mathcal{F}'(-h) Y_h = \Theta_h^{-1}, \quad \text{Res}_{z=-h} \mathcal{F}(z)^{-1} = Y_h \Theta_h Y_h^\dagger. \quad (\text{RM25})$$

The matrix mass remains through every collision. For distinct eigenvalues $h_i, h_j < \lambda_{\min} A$, their observed eigenvectors c_i, c_j satisfy the exact joint orthogonality identity

$$c_i^\dagger \left[I + B_H^\dagger (A - h_i I)^{-1} (A - h_j I)^{-1} B_H \right] c_j = \delta_{ij}. \quad (\text{RM26})$$

For equal eigenvalues this is RM24. For distinct ones it follows either by taking inner products of the reconstructed full eigenvectors or by the resolvent identity for the divided difference of $\mathcal{F}$. This establishes the full relation among poles and their observed vectors without an assumption that their observed Gram is diagonal.

6. Every slow eigenvalue in the energy graph

Keep the entire energy-minimizing section and its two Grams:

$$L_{\text{en}} b = \begin{pmatrix} -A^{-1} B_H b \\ b \end{pmatrix}, \quad Z = L_{\text{en}}^\dagger L_{\text{en}} = I + B_H^\dagger A^{-2} B_H, \quad S = L_{\text{en}}^\dagger H L_{\text{en}}.$$

Its actual isometry is $J_{\text{en}} = L_{\text{en}} Z^{-1/2}$, and its energy operator is

$$J_{\text{en}}^\dagger H J_{\text{en}} = Z^{-1/2} S Z^{-1/2}. \quad (\text{RM27})$$

Let h_j denote the eigenvalues of H in increasing order, including multiplicities, and ν_j those of RM27 in the same convention. Fix a computable $\alpha \leq \lambda_{\min} A$. For every j with $h_j < \alpha$,

$$h_j \leq \nu_j \leq \frac{h_j}{1 - h_j/\alpha}. \quad (\text{RM28})$$

There are at most $\dim B$ such eigenvalues, by the block inertia at a real number below α . The lower bound is the minimum principle on the isometric subspace $\text{im } J_{\text{en}}$. For the upper bound expand the exact resolvent in RM18:

$$\mathcal{F}(-h) = S - hZ - h^2 B_H^\dagger A^{-2} (A - hI)^{-1} B_H.$$

For $0 < h < \alpha$, the final positive matrix is at most $h^2(Z - I)/(\alpha - h) \preceq h^2 Z/(\alpha - h)$. Hence

$$S - \frac{h}{1 - h/\alpha} Z \preceq \mathcal{F}(-h). \quad (\text{RM29})$$

At $h = h_j$, block inertia says $\mathcal{F}(-h)$ has at least j nonpositive eigenvalues. The same is true of the left side of RM29. Conjugating by $Z^{-1/2}$ gives the upper bound. This proof includes every multiplicity and does not use differentiation of an individually labelled eigenvalue at a crossing.

In the proved native late cluster, $h_j = \exp[-2q \log(q/k) + O_h(q)]$ and $\alpha \geq e^{-o(q)}$. Thus RM28 gives a relative error of size at most

$$0 \leq \frac{\nu_j}{h_j} - 1 \leq \frac{h_j}{\alpha - h_j} = \exp[-2q \log(q/k) + O_{h,A,\mathcal{P}}(q)], \quad (\text{RM30})$$

with the same upper-bound interpretation as RM17, uniform over the admitted cluster. The original graph metric Z is retained. Its determinant cost is calculated for the canonical multiplication operator in the accompanying memory/energy proof;

RM28 alone does not assign that determinant cost for an arbitrary growing-degree filter.

Multiplying the proved bounds over all d slow directions also gives the entire cluster determinant, including all collisions:

$$0 \leq \sum_{j=1}^d \log \nu_j - \sum_{j=1}^d \log h_j \leq - \sum_{j=1}^d \log(1 - h_j/\alpha) \leq \frac{dh_+}{\alpha - h_+}. \quad (\text{RM31})$$

The last inequality follows by integrating $(1-t)^{-1}$ from zero to h_j/α . RF51 and RM23 make the final error exponentially small in the native family. Consequently RF54's full slow-energy return $-C_\partial dq + o(dq)$ is also the return of the first d eigenvalues of the actual energy-graph operator $Z^{-1/2}SZ^{-1/2}$. The four-cutoff error is bounded by the sum of the four displayed finite errors. This is an evaluated receiving result for the entire cluster; the full graph determinant still includes its remaining eigenvalues.

Proof sources and exact scope

The complete source-family memory and rational-filter identities are in the pinned 014 reader, specifically UH1-32 and IFE24-31. The harmonic and inverse decomposition used for the canonical native receiver are proved in HAR1-49 and ACT1-48. The root source polynomial, source mass, period and four-cutoff signs are unchanged.

Original-author TeX for Dusson-Sigal-Stamm was obtained intact from arXiv version 1 and read at the stated locators. The original Gamma and Meixner-Pollaczek sources remain attributed to their human authors in the rational-source proof. The estimates here are finite derivations, not an invocation of a source theorem without checking its hypotheses. Exact small-matrix checks and numerical norm checks have their separate fixtures and reported scope; they do not establish an asymptotic theorem.

Part V

Complete received memory and polynomial-filter continuation

The complete canonical memory term is below the kq scale, and polynomial filters have a determined slow spectrum

The new attachment provides the exact memory pencil, rational conductor map, and finite-rank decomposition of a filtered inverse. Those are the inputs used below. The current GitHub edition has incorporated the source-family and inverse-quotient papers; I read it at 2e84a4c23cacf7b00ec21b28fd18b189bccc87b9. The ES foundation remains at 49976b17491ba927df8bc7f2c2f81d39dfb15ce8.

There are two principal new conclusions.

First, the **entire canonical logarithmic memory contribution**, including arbitrarily small positive regularizers and its limit at zero, satisfies

$$\sup_{z \geq 0} g_N(z) = O_h(q \log(q/k)) = o_h(kq). \quad (1)$$

Thus it cannot carry the unresolved kq coefficient.

Second, let p_k be monic of degree d_k , with its poles in a fixed compact set disjoint from the original spectral lattice, allowing arbitrary repetitions and collisions. If

$$d_k \log(q+2) = o(q),$$

then $p_k(M_k)$ has **exactly d_k singular directions on the native late scale**

$$\exp[-q \log(q/k) + O_h(q)].$$

Their order- q exponent is evaluated below. The remaining singular values are bounded below by $\exp[-o(q)]$. The complete observed heat and resolvent at that late scale admit an explicitly constructed $d_k \times d_k$ model, with a finite error that is exponentially smaller than the retained observation fractions.

The finite arguments retain the original metrics. The evaluated order- q exponents use the same native monic-profile and reference zero-law inputs as the preceding accepted proof.

1. The canonical memory determinant is uniformly subleading

Fix an original cutoff and suppress N temporarily. All adjoints and orthogonal decompositions in this section use its actual G_N . Write

$$E = K \oplus K^\perp, \quad M = \begin{pmatrix} M_K & D \\ C & M_B \end{pmatrix},$$

and

$$H = M^\dagger M = \begin{pmatrix} A & B_H \\ B_H^\dagger & D_H \end{pmatrix}.$$

In particular,

$$D_H = D^\dagger D + M_B^\dagger M_B.$$

The attachment's complete pencil and logarithmic memory term are

$$\mathcal{F}(z) = zI + D_H - B_H^\dagger(zI + A)^{-1}B_H,$$

$$g(z) = \log \det(zI + A) + \log \det(zI + D_H) - \log \det(zI + H). \quad (2)$$

It proves $g(z) \geq 0$, complete monotonicity for $z > 0$, and

$$g(z) = \int_0^\infty e^{-zt} \frac{\mathcal{Q}(t)}{t} dt,$$

where

$$\mathcal{Q}(t) = \text{Tr } e^{-tH} - \text{Tr } e^{-tA} - \text{Tr } e^{-tD_H} \geq 0.$$

The full blocks, rather than $M_B^\dagger M_B$ alone, occur in these formulas.

A finite bound using only one large singular value

Suppose

$$s_1(M) \leq R, \quad s_j(M) \leq C_0 \quad (j \geq 2),$$

and M is invertible. Let $m = \dim K$, with $0 < m < q$. Then

$$\boxed{0 \leq g(z) \leq g(0) \leq 4 \log R + 2(q-2) \log C_0 - 2 \log |\det M|.} \quad (3)$$

To prove it, A is the Gram of M restricted to the orthonormal kernel frame. Its determinant is bounded by the product of the largest m squared singular values of M :

$$\det A \leq R^2 C_0^{2(m-1)}.$$

Likewise,

$$\det D_H \leq R^2 C_0^{2(q-m-1)}.$$

These product bounds follow directly from Cauchy-Binet in an eigenbasis of $M^\dagger M$: the squared minors of an orthonormal inclusion sum to one.

Since

$$\det H = |\det M|^2,$$

the two estimates give (3). The extension to $z = 0$ is legitimate because H , A , and D_H are positive definite. The cases $m = 0$ or $m = q$ have $g = 0$.

For the original native operator,

$$M_N = C_N + r_N \ell_N, \quad C_N^\dagger = C_N,$$

the retained source bounds give

$$R_N = \epsilon_N + B_h q, \quad C_0 = B_h q, \quad \epsilon_N = \|r_N \ell_N\|_{G_N}.$$

The rank-one correction gives the bound on all s_j , $j \geq 2$; it does not require orthogonality of the arithmetic eigenvectors. The source's profile is

$$\log \epsilon_N = q\psi(s_N) + O_h(k \log^2(q+2)).$$

Its full root determinant is

$$\log |\det M_k| = q \log k + \frac{q}{2} I(\delta, \gamma) + O_{\delta, \gamma}(k),$$

where

$$I(\delta, \gamma) = \log(\delta^2 + \gamma^2) - 3 + \frac{\delta}{\gamma} \arctan \frac{\gamma}{\delta} + \frac{\gamma}{\delta} \arctan \frac{\delta}{\gamma}.$$

These are the previously retained native quantities, not determinants in a constructed jet metric.

Substitution into (3) yields, uniformly at the original cutoffs,

$$\begin{aligned} g_N(z) &\leq 2q \log(q/k) \\ &\quad + q[4\psi(s_N) + 2 \log B_h - I(\delta, \gamma)] \\ &\quad + O_h(k \log^2(q+2) + \log q), \quad z \geq 0. \end{aligned} \tag{4}$$

This proves (1). In particular,

$$\sup_{z \geq 0} \frac{|\mathcal{R}g_N(z)|}{kq} \rightarrow 0. \tag{5}$$

The same statement permits separately specified regularizers at the four endpoints: the bound is pointwise nonnegative and uniform over all $z \geq 0$.

The time-domain consequence is also exact:

$$\int_0^\infty \frac{\mathcal{Q}_N(t)}{t} dt = g_N(0) = O_h(q \log(q/k)). \tag{6}$$

The complete memory is retained, including its long-time contribution. Its total logarithmic-time cost is now controlled below kq .

The remaining local leakage is a different term

Retain

$$\ell_N(z) = \log \det(zI + D_H) - \log \det(zI + M_B^\dagger M_B).$$

The attachment's exact comparison is

$$\log \det \mathcal{F}_N(z) - \log \det(zI + M_B^\dagger M_B) = \ell_N(z) - g_N(z). \tag{7}$$

The two nonnegative terms are not identified.

Both admit the sharper finite estimate

$$0 \leq \ell_N(z), g_N(z) \leq \log \left(1 + \frac{R_N^2}{z} \right) + (m-1) \log \left(1 + \frac{B_h^2 q^2}{z} \right), \quad z > 0. \tag{8}$$

For g_N , take the Schur complement on the other block and use

$$g_N(z) \leq \log \det(I + A/z).$$

For ℓ_N ,

$$\ell_N(z) \leq \log \det(I + DD^\dagger/z).$$

Each relevant rectangular restriction of M_N has at most one singular value exceeding $B_h q$. This proves (8).

Consequently the **canonical** comparison in (7), not only the joint-source comparison, is $o(kq)$ whenever

$$\log^+(1/z_k) = o(q). \quad (9)$$

At still smaller regularizers, the complete memory g_N remains subleading by (5); any possible leading discrepancy in (7) must come from the actual local leakage ℓ_N .

This sharpens the attachment's dimension-times-largest-norm bound without changing either operator.

2. A degree- d rational family has d native late singular directions

Write the infinite odd lattice as

$$\mathcal{L} = \{(2a+1)\gamma - i(2b+1)\delta : a, b \in \mathbb{Z}\}.$$

Fix a compact set

$$\mathcal{P} \subset \mathbb{C} \setminus \mathcal{L}.$$

For each k , let $p = p_k$ be monic of degree $d = d_k$, with every root in $\mathcal{P}$, repeated according to its full multiplicity. No separation between the roots of p is assumed. The same p_k is used at all four original cutoffs.

Set

$$U_p a = [a/p], \quad \deg a < d,$$

and retain

$$H_{p,N} = U_p^* G_N U_p, \quad H_{p,N}^B = U_p^* \Lambda^* Q_{B,N} \Lambda U_p. \quad (10)$$

The attachment proves the exact algebraic injectivity of the rational conductor on the guard $Jd - v \leq q'$, including translated-pole collisions. Its Laurent coefficient inverse and its full native receiving minimum are retained here.

Define

$$n_0 = 2 \left\lfloor \frac{N+1-q}{2} \right\rfloor, \quad T_{k,N} = \frac{1}{|Q_k(0)| \sqrt{K_{Q_k, n_0}(0, 0)}}, \quad (11)$$

where $K_{Q,n}$ is the degree- n reproducing kernel for the full measure

$$|Q(y)|^2 d\sigma(y), \quad d\sigma(y) = \frac{|\Gamma(1/4 + iy/2)|^2}{2\pi} dy.$$

The Gamma mass remains $\sqrt{2\pi}$.

The uniform rational norm theorem

If

$$d \log(q+2) = o(q),$$

there is a constant depending only on the fixed packet, conductor, and $\mathcal{P}$, such that

$$E_{k,d} = C_{h,A,\mathcal{P}}(k + Jd) \log(q + Jd + 2) = o(q)$$

and

$$\boxed{u_k T_{k,N}^2 e^{-E_{k,d}} I_d \preceq H_{p,N}^B \preceq H_{p,N} \preceq u_k T_{k,N}^2 e^{E_{k,d}} I_d.} \quad (12)$$

In particular,

$$\boxed{H_{p,N}^B \succeq e^{-2E_{k,d}} H_{p,N}.} \quad (13)$$

The estimate is uniform through collisions of the filter poles. It does not invert their pairwise differences.

Proof of the new uniformity

The proof extends the zero-pole argument by using polynomial remainders on a contour, rather than separated partial fractions.

Choose a bounded region Ω containing $\mathcal{P}$, whose boundary Γ has positive distance from both $\mathcal{P}$ and $\mathcal{L}$, and whose interior excludes $\mathcal{L}$. A large disk with finitely many small lattice disks removed gives such a region. Let

$$R = \max_{z \in \mathcal{P}} |z|, \quad R_\Gamma = \max_{z \in \Gamma} |z|, \quad \rho_\Gamma = \text{dist}(\Gamma, \mathcal{P}) > 0.$$

For a function f holomorphic on Ω , the complete Hermite remainder is

$$\boxed{\text{rem}_p f(z) = \frac{1}{2\pi i} \int_\Gamma \frac{f(w)}{p(w)} \frac{p(w) - p(z)}{w - z} dw.} \quad (14)$$

The oriented boundary includes all holes. Evaluating the formula and its derivatives at every root of p proves the full multiplicity statement.

If coefficient vectors use the original monomial basis, then

$$\boxed{\|\text{coeff rem}_p f\|_2 \leq \frac{|\Gamma|}{2\pi} \sqrt{d} \left(\frac{R + \max(1, R_\Gamma)}{\rho_\Gamma} \right)^d \sup_\Gamma |f|.} \quad (15)$$

Indeed $|p(w)| \geq \rho_\Gamma^d$, and each coefficient of $(p(w) - p(z))/(w - z)$ is bounded by $(R + \max(1, R_\Gamma))^d$.

There is no inverse root separation in (14)–(15).

The reference estimates needed next are the complete IVO9–IVO12 bounds already proved in the current reader. In particular,

$$\int y^2 |Qf|^2 d\sigma \geq 2^{-27} (\deg Q)^2 \int |Qf|^2 d\sigma$$

on its explicit degree and root-radius domain. This yields a hard gap for the relation-polynomial zeros, complex evaluation bounds, and a product bound for changing the kernel degree by h steps. The latter costs $O(h \log q)$, with the full polynomial retained.

Let

$$H_0(y) = \frac{K_{Q_k, n_0}(y, 0)}{K_{Q_k, n_0}(0, 0)}.$$

Then

$$H_0(0) = 1, \quad \|Q_k H_0\|_\sigma^2 = \mathsf{K}_{Q_k, n_0}(0, 0)^{-1}.$$

For sufficiently large k , H_0 has no zero in Ω , because all its nonzero zeros have modulus at least a fixed positive multiple of q .

Given $\deg a < d$, define

$$h(y) = -H_0(y) \operatorname{rem}_p \frac{a(y)}{Q_k(y)H_0(y)}.$$

Then

$$\boxed{f(y) = \frac{a(y) + Q_k(y)h(y)}{p(y)}} \quad (16)$$

is an actual polynomial representative of $U_p a$, of degree at most N .

On Γ , the paired root product gives

$$\log^+ \sup_{\Gamma} \left| \frac{Q_k}{Q_k(0)} \right| + \log^+ \sup_{\Gamma} \left| \frac{Q_k(0)}{Q_k} \right| = O_{\mathscr{P}, h}(\log(k+2)). \quad (17)$$

For the reciprocal bound, separate the finitely many lattice roots inside $|z| \leq 2R_\Gamma$, using their fixed distance from Γ . On the remaining pairs,

$$-\log |1 - z^2/\omega^2| \leq \frac{4}{3} \frac{|z|^2}{|\omega|^2},$$

and

$$\sum_{\omega/\pm} \frac{1}{|\omega|^2} \leq \frac{2}{\delta^2} \sum_{\substack{1 \leq j \leq k \\ j \text{ odd}}} \frac{1}{j}.$$

Polynomial division is bounded uniformly over these pole collisions. The Gamma generating function gives, for $|z| \leq R$,

$$\sqrt{\mathsf{K}_{\sigma, L}(z, z)} \leq \frac{e^{1+\pi R/4}}{\sqrt{\mathfrak{m}_\sigma}} (L+1)^{3/4} (2L+1)^{(R+1)/2}. \quad (18)$$

Together with

$$\|(y-z)f\|_\sigma \geq \frac{\|f\|_\sigma}{2b_{L+1}}, \quad b_L = \sqrt{2L+1} \sum_{j=1}^L \frac{1}{j},$$

this bounds division by one factor $y-z$. Applying those divisions successively bounds division by p by a product of d polynomial-in- N factors. The generating function used in (18) is the original Meixner-Pollaczek specialization. (DLMF)

Equations (14)-(18) applied to (16) give the upper bound in (12).

For the observed lower bound, use the centred conductor

$$\tilde{\mathcal{T}}_A f(y) = \sum_j a_j f(y + \tau_j), \quad \tau_j = (b_j - 4)/i.$$

Its first nonzero symbol moment is $i^{-v} \mu_v$. Put

$$\mathcal{B}_p(y) = \prod_{j=1}^J p(y + \tau_j), \quad L = Jd,$$

and

$$N_a(y) = \mathcal{B}_p(y) \tilde{\mathcal{T}}_A(a/p)(y).$$

The complete numerator has degree at most $L - v - 1$. Its coefficient map

$$a \longmapsto N_a$$

has an inverse bound

$$\|\text{coeff } a\|_2 \leq \mathcal{W}_d \|\text{coeff } N_a\|_2, \quad \log^+ \mathcal{W}_d = O_{A, \mathcal{P}}(d \log(d+2)). \quad (19)$$

Here is an explicit construction of that bound. Write

$$e(z) = z^{-v} \tilde{E}_A(-z), \quad |[z^j]e(z)^{-1}| \leq H_* \varrho^{-j},$$

where $\varrho > 0$ is chosen so that

$$|e(z) - e(0)| \leq |e(0)|/2 \quad (|z| \leq \varrho), \quad H_* = 2v!/|\mu_v|.$$

Let $R_B = R + \max_j |\tau_j|$, and set

$$B_{d,L} = \sqrt{L} \sum_{j=0}^{v+d-1} \binom{L+j-1}{j} R_B^j,$$

$$A_d = \max_{1 \leq n \leq d} (n-1)! \sum_{\ell=1}^n \frac{H_* \varrho^{-(n-\ell)}}{(v+\ell-1)!}.$$

Then one may take

$$\boxed{\mathcal{W}_d = \sqrt{d}(1+R)^d A_d B_{d,L}.} \quad (20)$$

The first factor recovers the numerator from its Laurent coefficients; $B_{d,L}$ bounds the Laurent coefficients of $N_a/\mathcal{B}_p$; A_d inverts the original symbol convolution. This is the attachment's full Laurent inverse, not inversion of a selected residue.

Any lower polynomial representative f_- satisfies

$$\mathcal{B}_p f_- - N_a = Q_- h_-. \quad (21)$$

Taking the full remainder modulo $\mathcal{B}_p$, and applying (14) on a circle containing its roots, bounds every numerator coefficient by a constant times

$$|Q_-(0)| \sqrt{\mathcal{K}_{Q_-, N-v+Jd-q_-}(0,0)} \|Q_- h_-\|_\sigma.$$

The nearby-degree kernel comparison and the exact factorization

$$Q_k = Q_- D_\partial, \quad \deg D_\partial = 16k - 48,$$

compare this factor to $T_{k,N}^{-1}$, with logarithmic cost

$$O_{h,A,\mathcal{P}}((k+Jd) \log(q+Jd+2)).$$

The lower root product and every boundary factor are retained. The numerator term in (21) has only

$$\exp[O_{A,\mathcal{P}}(Jd \log(Jd+2))]$$

polynomial norm. Meanwhile the explicit IVO23 estimate gives

$$\log T_{k,N} \geq q \log(q/k) - O_h(q).$$

It therefore absorbs that numerator error on a finite eventual domain. Finally, minimize through the actual conductor and observation fibres:

$$\|P_{B,N} U_p a\|_{G_N} \geq \frac{\sqrt{\ell_{k,N}}}{\mathcal{F}_N} \|\tilde{T}_A(a/p)\|_{\sigma, Q_-, N-v},$$

with the original $\mathcal{F}_N$. This proves the lower bound in (12).

The proof neither integrates a/p through a pole on the real line nor replaces the original observation kernel by the conductor kernel.

Why the finite-rank part actually has rank d

The attachment correctly warns that the finite-rank covariance can be singular before its guard is checked. Here its positivity is proved on the growing native family. Let

$$T_p = \text{rem}_p L_N.$$

For a polynomial a of degree below d ,

$$L_N[a] = a - Q_k h_a.$$

The second term is the complete relation projection. Applying the contour estimate to it gives

$$\boxed{T_p[a] = (I - \mathcal{E}_{p,N})a, \quad \|\mathcal{E}_{p,N}\| \leq T_{k,N}^{-1} e^{C_{h,A,\mathcal{D}}(k+d) \log(q+d+2)}. \quad (22)}$$

The right side tends to zero. Thus T_p is onto for sufficiently large k , uniformly over the allowed polynomial family.

Now retain the exact division identity

$$\boxed{p(M)^{-1} = B_p + U_p T_p, \quad B_p = J_N \mathfrak{Q}_p L_N, \quad (23)}$$

where $\mathfrak{Q}_p$ is ordinary polynomial quotient division. Its native norm obeys

$$\log^+ \|B_p\| = O_{h,\mathcal{D}}(k + d \log(q+2)).$$

Equation (22) bounds the smallest singular value of T_p through its explicit right inverse on the low polynomial classes. Together with (12), it gives

$$T_{k,N} e^{-E_{k,d}} \leq s_d(U_p T_p) \leq s_1(U_p T_p) \leq T_{k,N} e^{E_{k,d}},$$

after increasing the displayed constant.

Since $T_{k,N} \geq \exp[q \log(q/k) - O_h(q)]$, the rank- d term dominates B_p . Weyl's singular-value inequalities prove

$$\boxed{\log s_{q-d+j,N}(p(M)) = -\log T_{k,N} + O(E_{k,d}), \quad 1 \leq j \leq d, \quad (24)}$$

uniformly in j , while

$$\boxed{s_{q-d,N}(p(M)) \geq e^{-O(E_{k,d})}. \quad (25)}$$

This proves the asserted count of directions on the native late scale.

The separation result itself uses the finite source comparison and the explicit Gamma estimates. The next order- q coefficient uses the additional profile and zero-law inputs.

3. Every one of those d exponents is evaluated

The preceding reference-kernel calculation gives

$$\log T_{k,N} = q \log(q/k) + q \left[\psi(s) + J(s) - \frac{I(\delta, \gamma)}{2} \right] + o_h(q) \quad (26)$$

when $(N + 1 - q)/q \rightarrow s$, with the same adjacent-cutoff convention as before. The logarithmic relation limit is justified by its finite hard gap, not by weak convergence alone.

Therefore

$$\boxed{\log s_{q-d+j,N}(p_k(M_k)) = -q \log(q/k) + q \left[\frac{I}{2} - \psi(s) - J(s) \right] + o_h(q), \quad 1 \leq j \leq d.} \quad (27)$$

The remainder is uniform over the whole admissible pole family and all j .

At $N = q - 1, q$, it is explicitly

$$O_{h,A,\mathcal{P}}(k \log^2(q+2) + Jd \log(q + Jd + 2)).$$

No effective convergence threshold for the upper reference zero law is added.

Let

$$\mathfrak{C} = 4[\psi(0) - \psi(1) - J(1) - 1].$$

The retained interval certificate gives

$$1.353893 < \mathfrak{C} < 1.354751.$$

Consequently

$$\boxed{\frac{1}{dq} \mathcal{R} \log \det H_{p,N}^B \longrightarrow \mathfrak{C},} \quad (28)$$

and

$$\boxed{\frac{1}{dq} \mathcal{R} \sum_{j=q-d+1}^q \log s_{j,N}(p(M))^2 \longrightarrow -\mathfrak{C}.} \quad (29)$$

The numerical enclosure is inherited from the supplied certificate, not newly recomputed.

For

$$\tau_k^* = \exp \left\{ 2q \left[\log(q/k) - 1 - \frac{I(\delta, \gamma)}{2} \right] \right\},$$

the complete filtered positive heat satisfies

$$\boxed{\frac{1}{d} \mathcal{R} \operatorname{Tr} e^{-\tau_k^* p(M)^\dagger p(M)} \longrightarrow 2.} \quad (30)$$

The same physical time is used at all four cutoffs.

The original observation preserves all d slow directions up to a factor $e^{-o(q)}$. Indeed, the angle between their subspace and $\operatorname{im} U_p$ is bounded by

$$\frac{\|B_p\|}{s_d(p(M)^{-1})},$$

which is exponentially smaller than the square root of the observation bound (13). Applying the sharp projection inequality from the new GitHub proof gives

$$\boxed{\mathcal{R} \operatorname{Tr} \left(\Lambda e^{-\tau_k^* p(M)^\dagger p(M)} L_{B,N} \right) > 0} \quad (31)$$

eventually, and

$$\boxed{\frac{1}{q} \log \left[\frac{1}{d} \mathcal{R} \operatorname{Tr} \left(\Lambda e^{-\tau_k^* p(M)^\dagger p(M)} L_{B,N} \right) \right] \rightarrow 0.} \quad (32)$$

The projection inequality's sharp coefficient, including its $\sqrt{1-\ell}$ term, is retained.

For a nonmonic filter $a_k p_k$, the exact restoration is

$$s_j(a_k p_k(M)) = |a_k| s_j(p_k(M)), \quad \tau \mapsto |a_k|^2 \tau.$$

No leading coefficient is silently normalized out of the heat calculation.

4. A $d \times d$ model calculates the complete late observed heat and resolvent

The preceding theorem supplies the asymptotic separation. The following finite statement supplies a directly computable model.

Keep the actual matrices

$$H_p = U_p^* G U_p, \quad T_p = \operatorname{rem}_p L_N,$$

and define

$$\boxed{K_p = H_p^{1/2} T_p G^{-1} T_p^* H_p^{1/2}, \quad Y_p = \Lambda U_p H_p^{-1/2}.} \quad (33)$$

Its observation Gram is

$$\boxed{\Theta_p = Y_p^\dagger Y_p = H_p^{-1/2} H_p^B H_p^{-1/2}.} \quad (34)$$

Neither K_p nor Θ_p is assumed diagonal, and they need not commute.

Let

$$\beta \geq \|B_p\|, \quad \kappa_- = \lambda_{\min} K_p, \quad \kappa_+ = \|K_p\|.$$

The finite guard

$$\kappa_- > 4\beta^2 \quad (35)$$

ensures the separated d -dimensional cluster. The native estimates above prove this guard eventually.

The complete observed resolvent

For every $z \geq 0$,

$$\boxed{\left\| \Lambda(zI + p(M)^\dagger p(M))^{-1} L_{B,N} - Y_p(\mathbf{K}_p^{-1} + zI)^{-1} Y_p^\dagger \right\|} \leq \varepsilon_R := 2\beta\sqrt{\kappa_+} + \beta^2. \quad (36)$$

To prove it, put $X = p(M)^{-1}$ and $F = U_p T_p$. Equation (23) gives

$$\|XX^\dagger - FF^\dagger\| \leq 2\beta\|F\| + \beta^2.$$

The isometry $V = U_p H_p^{-1/2}$ satisfies

$$FF^\dagger = V\mathbf{K}_p V^\dagger.$$

For positive A, B ,

$$A(I + zA)^{-1} - B(I + zB)^{-1} = (I + zA)^{-1}(A - B)(I + zB)^{-1}.$$

Both outer factors are contractions. Apply this identity to

$$A = XX^\dagger, \quad B = FF^\dagger,$$

and then the original contraction Λ and isometry $L_{B,N}$. This proves (36).

The complete observed heat

For every $\tau > 0$,

$$\boxed{\left\| \Lambda e^{-\tau p(M)^\dagger p(M)} L_{B,N} - Y_p e^{-\tau \mathbf{K}_p^{-1}} Y_p^\dagger \right\|_1} \leq \frac{4}{e^{2\tau}} \left[2\sqrt{q} \beta \sqrt{\text{Tr } \mathbf{K}_p} + q\beta^2 \right]. \quad (37)$$

The proof is finite. Define

$$f_\tau(x) = \begin{cases} e^{-\tau/x}, & x > 0, \\ 0, & x = 0. \end{cases}$$

Its scalar Lipschitz constant is

$$\sup_{x \geq 0} |f'_\tau(x)| = \frac{4}{e^{2\tau}}.$$

For positive Hermitian A, B , use their two eigenbases:

$$\langle u_i, (f_\tau(A) - f_\tau(B))v_j \rangle = [f_\tau(a_i) - f_\tau(b_j)] \langle u_i, v_j \rangle.$$

Squaring and summing proves

$$\|f_\tau(A) - f_\tau(B)\|_{\text{HS}} \leq \frac{4}{e^{2\tau}} \|A - B\|_{\text{HS}}.$$

Moreover,

$$\|XX^\dagger - FF^\dagger\|_{\text{HS}} \leq 2\beta\sqrt{\text{Tr } \mathbf{K}_p} + \sqrt{q}\beta^2.$$

Trace norm is at most $\sqrt{q}$ times Hilbert-Schmidt norm. The quotient contraction preserves the resulting trace-norm bound, giving (37).

Thus the observed scalar heat is calculated by

$$\boxed{\text{Tr} \left(\Theta_p e^{-\tau \mathbf{K}_p^{-1}} \right)} \quad (38)$$

with the explicit error in (37). The product is retained in its given order; no commuting approximation is used.

At the native late scale,

$$\tau = \exp[2q \log(q/k) + O_h(q)],$$

the error in (37) is

$$\exp[-q \log(q/k) + O_h(q)],$$

up to the explicitly displayed polynomial factors. It is smaller than the retained observation fractions $e^{-o(q)}$. This is substantially stronger at that scale than the attachment's uniform $O(m)$ trace comparison.

The corresponding memory pencil

Let $a = \lambda_{\min} \Theta_p > 0$, and use the original isometry

$$V_B = Y_p \Theta_p^{-1/2}$$

onto the observed rational subspace. Define

$$\mathcal{F}_{\text{red}}(z) = \left[V_B^\dagger \Lambda(zI + p(M)^\dagger p(M))^{-1} L_{B,N} V_B \right]^{-1}.$$

This is the exact Schur complement after eliminating both the original kernel and the remaining observed complement.

Its explicit model is

$$\boxed{\widehat{\mathcal{F}}_{\text{red}}(z) = \Theta_p^{-1/2} (\mathbf{K}_p^{-1} + zI) \Theta_p^{-1/2}.} \quad (39)$$

If

$$\delta_R(z) = \frac{\varepsilon_R(\kappa_-^{-1} + z)}{a} < 1,$$

then (36) gives

$$\boxed{\frac{\widehat{\mathcal{F}}_{\text{red}}(z)}{1 + \delta_R(z)} \preceq \mathcal{F}_{\text{red}}(z) \preceq \frac{\widehat{\mathcal{F}}_{\text{red}}(z)}{1 - \delta_R(z)}}. \quad (40)$$

For $z = x/T_{k,N}^2$ with bounded $x \geq 0$,

$$\delta_R(z) = \exp[-q \log(q/k) + O_h(q)].$$

The slope Θ_p^{-1} in (39) is the actual observed mass matrix. Setting it equal to the identity would change the pole locations relative to the static Schur matrix.

5. The static Schur eigenvalue and the true slow pole are quantitatively different

This can be stated without a polynomial model.

Let $H > 0$ have one simple eigenvalue h , with unit eigenvector w , and assume all other eigenvalues are at least $g > h$. Let

$$\theta = \|P_B w\|^2 > 0.$$

In the original kernel/observation decomposition,

$$H = \begin{pmatrix} A & B_H \\ B_H^\dagger & D_H \end{pmatrix}.$$

For a unit $x \in K$,

$$|\langle w, x \rangle|^2 \leq 1 - \theta.$$

Therefore

$$\boxed{A \succeq [h(1 - \theta) + g\theta]I \succeq g\theta I.} \quad (41)$$

The kernel's memory frequencies are separated from the late eigenvalue by this finite bound.

Let

$$S = D_H - B_H^\dagger A^{-1} B_H = \mathcal{F}(0).$$

Since

$$S^{-1} = \Lambda H^{-1} L_{B,N},$$

its spectral decomposition gives

$$S^{-1} = \frac{\theta}{h} b b^\dagger + R, \quad b = \frac{\Lambda w}{\sqrt{\theta}}, \quad 0 \preceq R \preceq g^{-1} I.$$

Hence

$$\boxed{\frac{h}{\theta + h/g} \leq \lambda_{\min}(S) \leq \frac{h}{\theta}, \quad \lambda_2(S) \geq g.} \quad (42)$$

The true pole is at $-h$, not generally at $-\lambda_{\min}(S)$. Indeed,

$$\boxed{\mathcal{F}(-h)b = 0, \quad b^\dagger \mathcal{F}'(-h)b = \frac{1}{\theta},} \quad (43)$$

and

$$\boxed{\operatorname{Res}_{z=-h} \mathcal{F}(z)^{-1} = \theta b b^\dagger.} \quad (44)$$

The full eigenvector is reconstructed by

$$w = \sqrt{\theta} \begin{pmatrix} -(A - hI)^{-1} B_H b \\ b \end{pmatrix}.$$

Its norm is exactly the derivative identity in (43). These are finite Feshbach-Schur identities, with the original metric and residue retained; the general method

has standard precedents, while the displayed bounds specify this receiver (Dusson et al., 2021). (EMS Press)

For the native $M_k^\dagger M_k$, the inverse-power observation theorem and the source's singular-line estimate give

$$\theta \geq e^{-O_{h,A}(k \log q)}, \quad g \geq e^{-O_h(k + \log q)}.$$

Thus $\lambda_{\min}(S)$ has the same order- q logarithmic asymptotic as $h = s_{\min}(M_k)^2$, but its finer scale retains θ .

A complete energy-graph metric, with its cost retained

Put

$$Z = I + B_H^\dagger A^{-2} B_H.$$

The energy-minimizing section

$$L_{\text{en}} b = \begin{pmatrix} -A^{-1} B_H b \\ b \end{pmatrix}$$

has original norm Gram Z , and energy Gram S . Thus

$$J_{\text{en}} = L_{\text{en}} Z^{-1/2}$$

is an actual original-metric isometry, and

$$J_{\text{en}}^\dagger H J_{\text{en}} = Z^{-1/2} S Z^{-1/2}. \quad (45)$$

If $\alpha \leq \lambda_{\min} A$ and $h < \alpha$, its smallest eigenvalue ν satisfies

$$\boxed{h \leq \nu \leq \frac{h}{1 - h/\alpha}}. \quad (46)$$

To prove it, expand the exact pencil:

$$\mathcal{F}(z) = S + zZ - z^2 B_H^\dagger A^{-2} (A + zI)^{-1} B_H.$$

At $z = -h$, the remainder is positive and bounded by

$$\frac{h^2}{\alpha - h} Z.$$

Schur inertia, or the minimum principle applied at the zero of $\mathcal{F}(-h)$, gives (46). The same argument applies to every slow eigenvalue below α , including multiplicities.

For the native rank-one correction, this graph's entire determinant cost is also subleading. Let

$$\alpha = g\theta, \quad R_N = \epsilon_N + B_h q, \quad C_0 = B_h q.$$

Then

$$m \log \alpha \leq \log \det A \leq 2 \log R_N + 2(m-1) \log C_0. \quad (47)$$

Also

$$H - C_N^2$$

has rank at most two. Therefore all but at most two singular values of $A^{-1}B_H$ are at most C_0^2/α , while all are at most R_N^2/α . Consequently

$$\log \det Z \leq 2 \log \left(1 + \frac{R_N^4}{\alpha^2} \right) + (m-2) \log \left(1 + \frac{C_0^4}{\alpha^2} \right). \quad (48)$$

On the original family,

$$|\log \det A| + \log \det Z = O_{h,A}(q \log q) = o(kq).$$

Define the full energy quotient in the original target coefficients by

$$\mathcal{E}_{B,N} = [\Lambda(M_k^* G_N M_k)^{-1} \Lambda^*]^{-1}.$$

The exact determinant factorization is

$$\det(Q_{B,N}^{-1/2} \mathcal{E}_{B,N} Q_{B,N}^{-1/2}) = \frac{|\det M_k|^2}{\det A}.$$

The root determinant is independent of cutoff. Hence

$$\mathcal{R} \log \det(Q_{B,N}^{-1/2} \mathcal{E}_{B,N} Q_{B,N}^{-1/2}) = o_{h,A}(kq). \quad (49)$$

The same leading zero holds for the energy-graph operator in (45), because (48) retains its additional metric determinant.

This is an evaluated kq -scale coefficient for the stated energy quotient. It is not a substitution for the separate proper-source F_R, J_* minimum.

6. An explicit ES family now produces a kq -scale native determinant and a $4k$ -dimensional slow sector

Use the actual witness

$$\frac{4}{13} = \frac{1}{4} + \frac{1}{18} + \frac{1}{468}.$$

Its literal prime-denominator polynomial is

$$P_{\text{ES}}(t) = t^4 - 503t^3 + 16738t^2 - 168480t + 438048. \quad (50)$$

The roots are exactly 13, 4, 18, 468. Its reciprocal trace is

$$\frac{168480}{438048} = \frac{5}{13}.$$

Choose the explicit growing filter

$$p_k(y) = P_{\text{ES}}(y)^k, \quad d_k = 4k. \quad (51)$$

Its real poles are uniformly separated from the original nonreal odd lattice by at least δ . Every repeated pole is retained. The degree condition $4k \log q = o(q)$ holds.

Therefore $P_{\text{ES}}(M_k)^k$ has exactly $4k$ directions on the native late scale, with the exponent in (27), and

$$\mathcal{R} \log \det [U_{p_k}^* \Lambda^* Q_{B,N} \Lambda U_{p_k}] = 4\mathfrak{C} kq + o(kq). \quad (52)$$

The inherited numerical enclosure gives

$$\boxed{5.415572 < 4\mathfrak{C} < 5.419004.} \quad (53)$$

At the common time τ_k^* ,

$$\boxed{\frac{1}{4k} \mathcal{R} \text{Tr} e^{-\tau_k^* p_k (M_k)^\dagger p_k (M_k)} \longrightarrow 2.} \quad (54)$$

The actual measured heat has the eventual positive sign and exponential-rate statement (31)–(32).

The algebraic conductor guard is explicit:

$$J(4k) - v \leq (k - 7)^2.$$

Since $J \leq 81$, the sufficient condition $k \geq 341$, $k \equiv 1 \pmod{4}$, ensures this guard. The additional analytic eventual guards are those in the norm proof; this algebraic threshold is not presented as the asymptotic threshold.

The ES action map and its exact defect

Let

$$\mathcal{A}_{p_k} = \mathbb{C}[t]/(p_k),$$

with its full multiplicities, and let C_{p_k} be multiplication by t in the original coefficient frame. The map

$$U_{p_k} : \mathcal{A}_{p_k} \rightarrow E_k, \quad a \mapsto [a/p_k]$$

is injective on the stated domain. It is a linear embedding, not an algebra homomorphism.

Its action defect is exactly

$$\boxed{M_k U_{p_k} = U_{p_k} C_{p_k} + [1] e_{d_k-1}^*} \quad (55)$$

Indeed, division of $ta(t)$ by the monic p_k leaves the scalar quotient equal to the leading coefficient of a .

In the physical S -coordinate,

$$\boxed{M_S U_{p_k} = U_{p_k} (kI/2 + iC_{p_k}) + i[1] e_{d_k-1}^*} \quad (56)$$

The centred coordinate and the factor i have not been suppressed. Composing with the existing physical unit multiplication transports the defect vector by that same unit.

The attained compression to this rational space is

$$U_{p_k}^{\dagger G_N} M_k U_{p_k} = C_{p_k} + H_{p_k, N}^{-1} U_{p_k}^* G_N [1] e_{d_k-1}^* \quad (57)$$

Thus the original metric correction is fully specified.

The ES integers here are filter poles, not asserted zeta zeros. Equations (55)–(57), rather than equality of the two spectral actions, are the explicit connection.

The original kernel coefficient is preserved through this rational quotient

For any admissible p , define

$$\pi_p[x] = \mathfrak{Q}_p(\text{rem}_{Q_k}(p x)) \in \mathcal{P}_{<q-d}.$$

Then

$$\ker \pi_p = \text{im } U_p,$$

and the low-polynomial inclusion is a right inverse.

The original kernel remains injective under π_p , and its metric loss is exactly the common angle determinant

$$\begin{aligned} \delta_{p,N} &= \log \det H_{p,N} - \log \det H_{p,N}^B \\ &= \log \det(I_K^* G_N I_K) - \log \det[(\pi_p I_K)^* (\pi_p G_N^{-1} \pi_p^*)^{-1} (\pi_p I_K)]. \end{aligned}$$

By (13), only $\min(m, d)$ nontrivial angle eigenvalues contribute:

$$0 \leq \delta_{p,N} \leq 2 \min(m, d) E_{k,d} = o(kq).$$

Consequently

$$\boxed{\mathcal{R} \log \det(I_K^* G_N I_K) = \mathcal{R} \log \det[(\pi_p I_K)^* (\pi_p G_N^{-1} \pi_p^*)^{-1} (\pi_p I_K)] + o(kq).} \quad (58)$$

This now holds for the complete colliding-pole family, including (51), rather than only powers of y^{-1} .

The finite $d \times d$ model (33)–(40), the quotient (58), and the explicit action defect (55) provide the common receiver through which this ES family enters the native programme.

Consequence for the remaining calculation

The attachment's full memory term is now quantitatively removed from the kq coefficient by (1)–(8), including its zero-regularizer limit. The rational-filter argument determines all late singular directions in growing degree and replaces their full measured heat by the actual matrices K_p, Θ_p , with the complete error (37). The original kernel can be quotiented by that entire rational sector without changing its leading coefficient, while the removed observed determinant has the evaluated coefficient (28).

The remaining original invariant-kernel coefficient and independent proper-source target still have their own matrices. Equations (49) and (58) specify which new replacements preserve their relevant scales; they do not assign values to unrelated minima or to the complex projected-current numerator.

The new results here are analytic derivations. Container execution remained unavailable, so I am not reporting a fresh numerical or symbolic test receipt, and neither repository was modified.

References

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